# AWS VPC PEERING
# STUDENT PROJECT REPORT

| Cover detail | Fill before submission |
|---|---|
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| Email ID | engineeromesh@gmail.com |
| Batch / Cohort | 1 |
| Project title | AWS VPC Peering: Private Connectivity Between VPC A and VPC B |
| AWS Region | Asia Pacific (Sydney) — ap-southeast-2 |
| Completion date | October 27, 2026 |

**Purpose**

This report documents the VPC peering design, routing, security rules and connectivity verification for two VPCs in AWS.

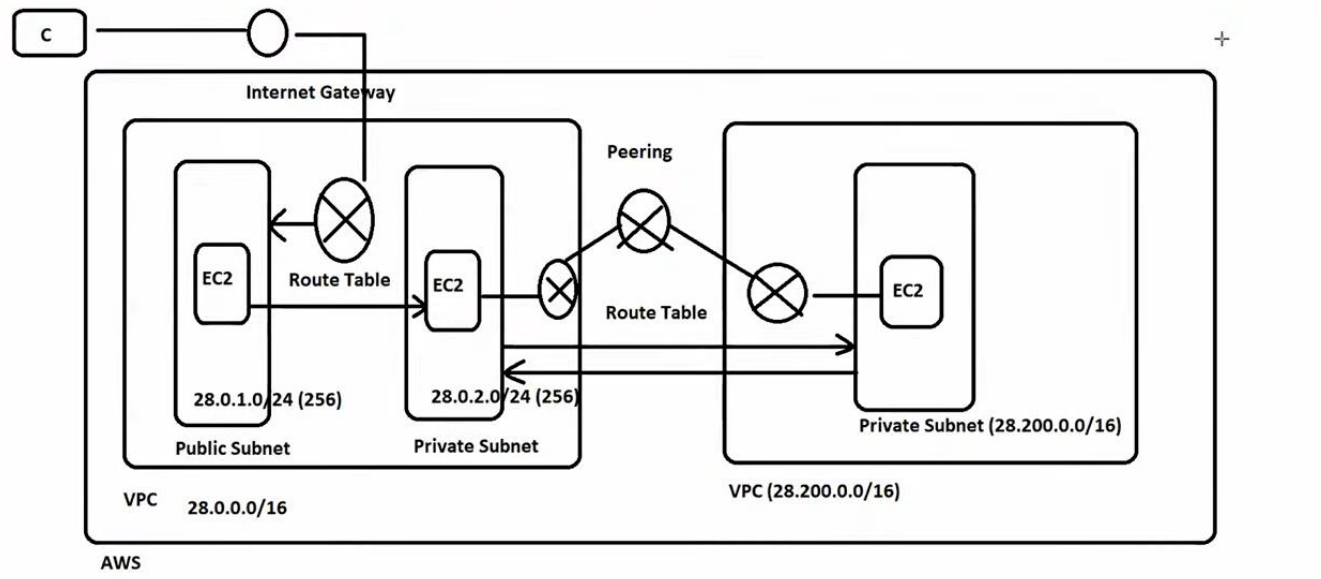
## 1. Architecture and Resource Inventory

The project uses VPC peering to allow private communication between App-A in VPC A and App-B in VPC B. A bastion host in the public subnet of VPC A provides SSH access to the private instances.

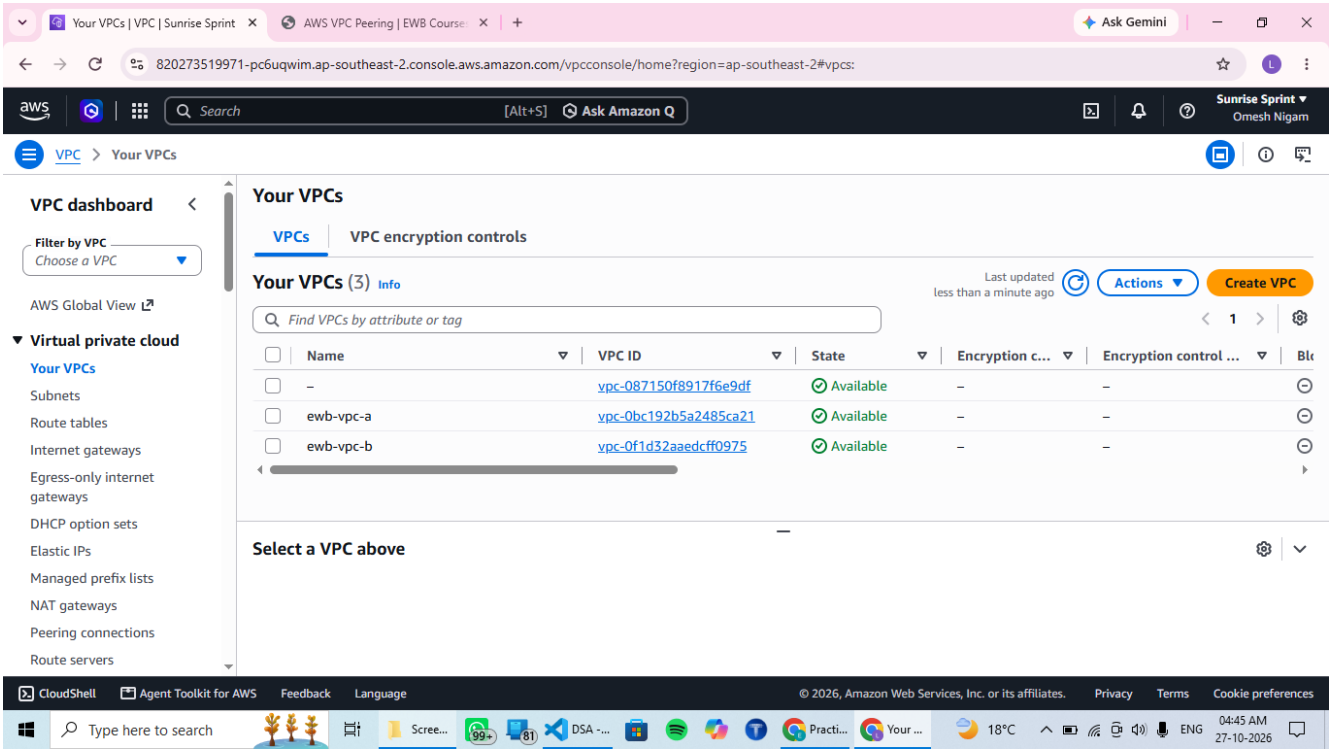
**Logical path**: Laptop → Bastion (public IPv4 13.55.162.231) → App-A (10.10.2.10) or App-B (10.20.1.10). App-A and App-B communicate over the peering connection using private IP addresses.

| Resource | Name — Resource ID | CIDR / IP | Purpose / placement |
|---|---|---|---|
| VPC A | ewb-vpc-a — vpc-0bc192b5a2485ca21 | 10.10.0.0/16 | VPC containing bastion and App-A |
| VPC B | ewb-vpc-b — vpc-0f1d32aaedcff0975 | 10.20.0.0/16 | VPC containing App-B |
| Public subnet A | ewb-a-public — subnet-07f524e6c4cba75d3 | 10.10.1.0/24 | VPC A; bastion subnet |
| Private subnet A | ewb-a-private — subnet-0f71c85ec2224a6f0 | 10.10.2.0/24 | VPC A; App-A subnet |
| Private subnet B | ewb-b-private — subnet-0373382ac48e99d7b | 10.20.1.0/24 | VPC B; App-B subnet |
| Bastion | ewb-bastion — i-035ba43fca2577fde | Private: 10.10.1.10<br>Public: 13.55.162.231 | Public subnet A; SSH jump host |
| EC2 App-A | ewb-app-a — i-0c0b922f6d8095c43 | Private: 10.10.2.10; no public IPv4 | Private subnet A |
| EC2 App-B | ewb-app-b — i-031b78f04d1f47db1 | Private: 10.20.1.10; no public IPv4 | Private subnet B |
| VPC peering | pcx-0a3719dd486418f19 | VPC A ↔ VPC B | Peering connection; verify state is Active |

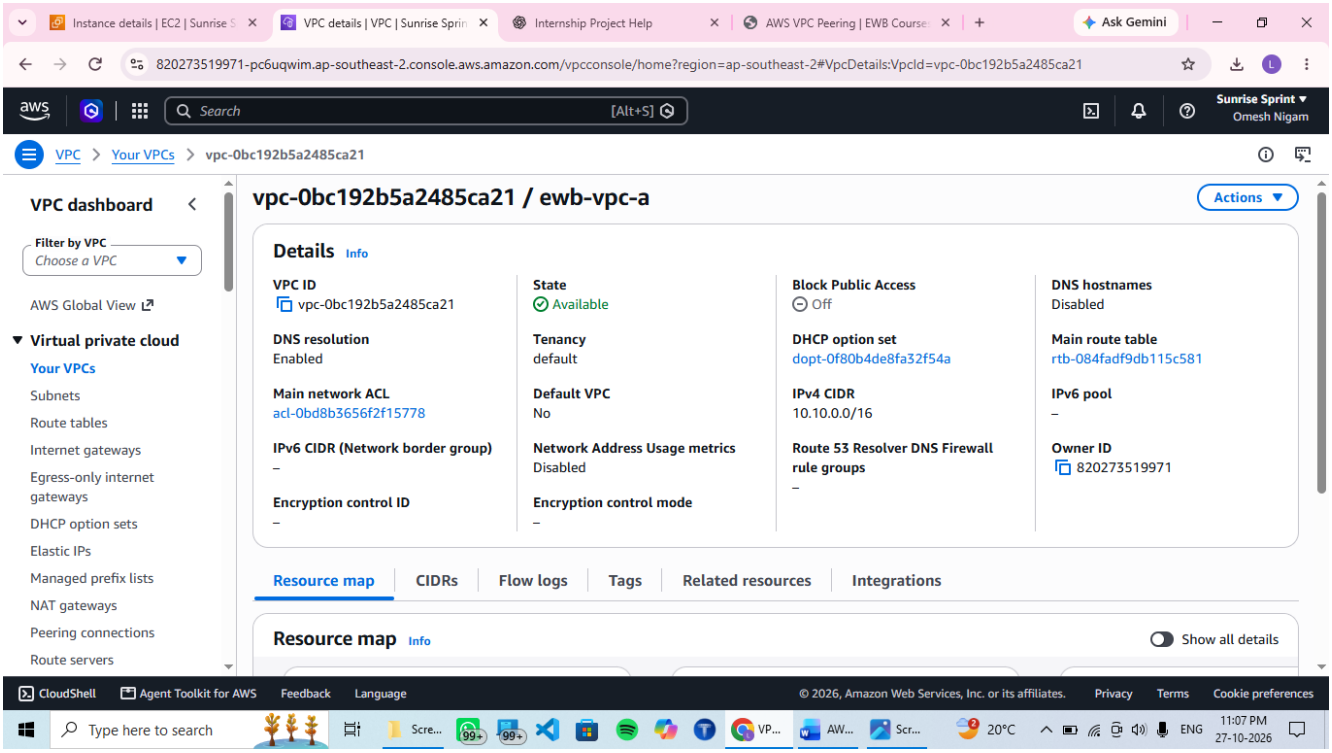
## [SCREENSHOT 1: Architecture Diagram]

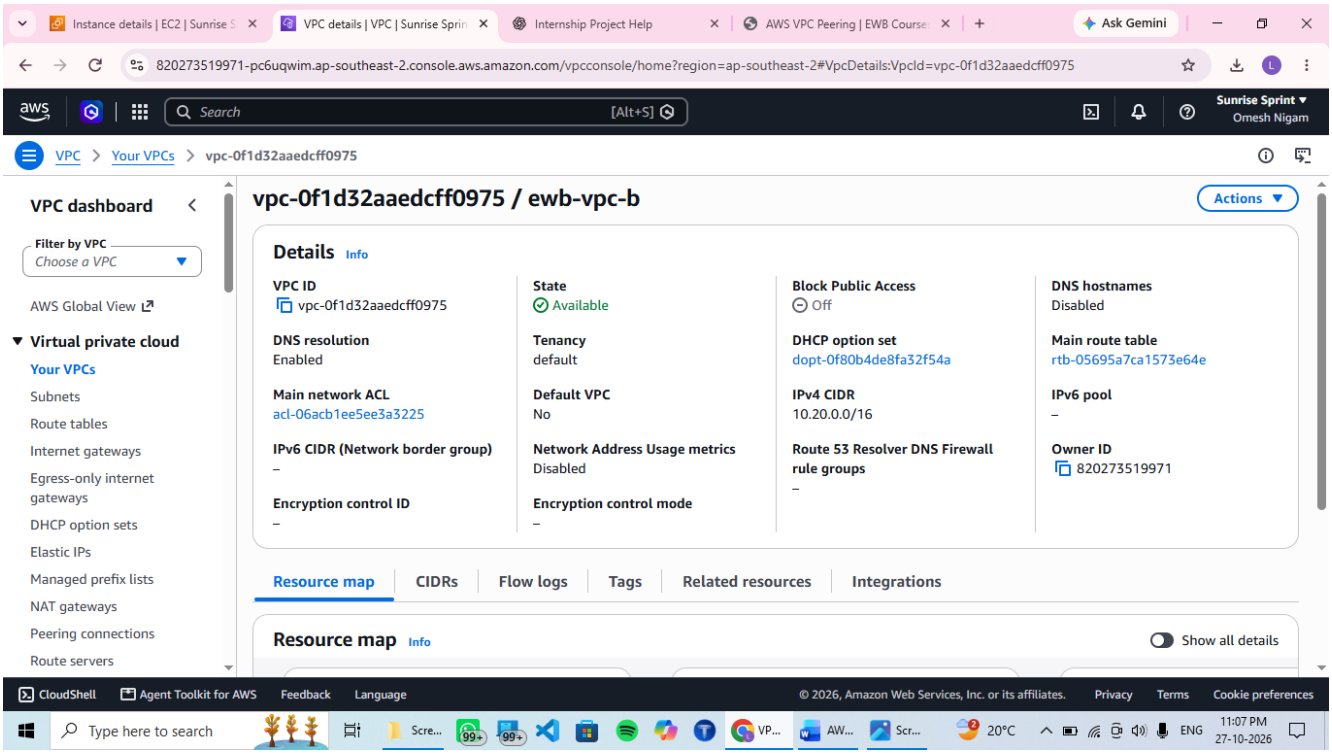



## [SCREENSHOT 2: Showing all VPCs]

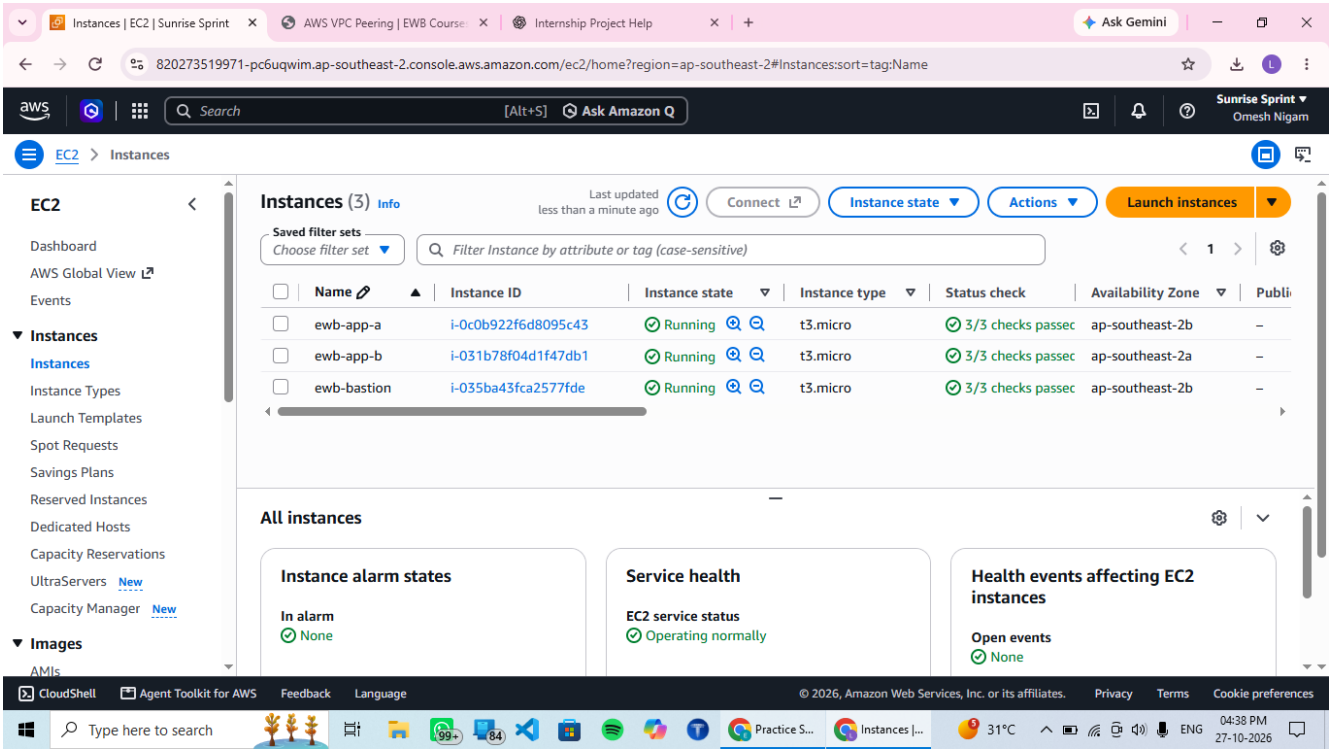


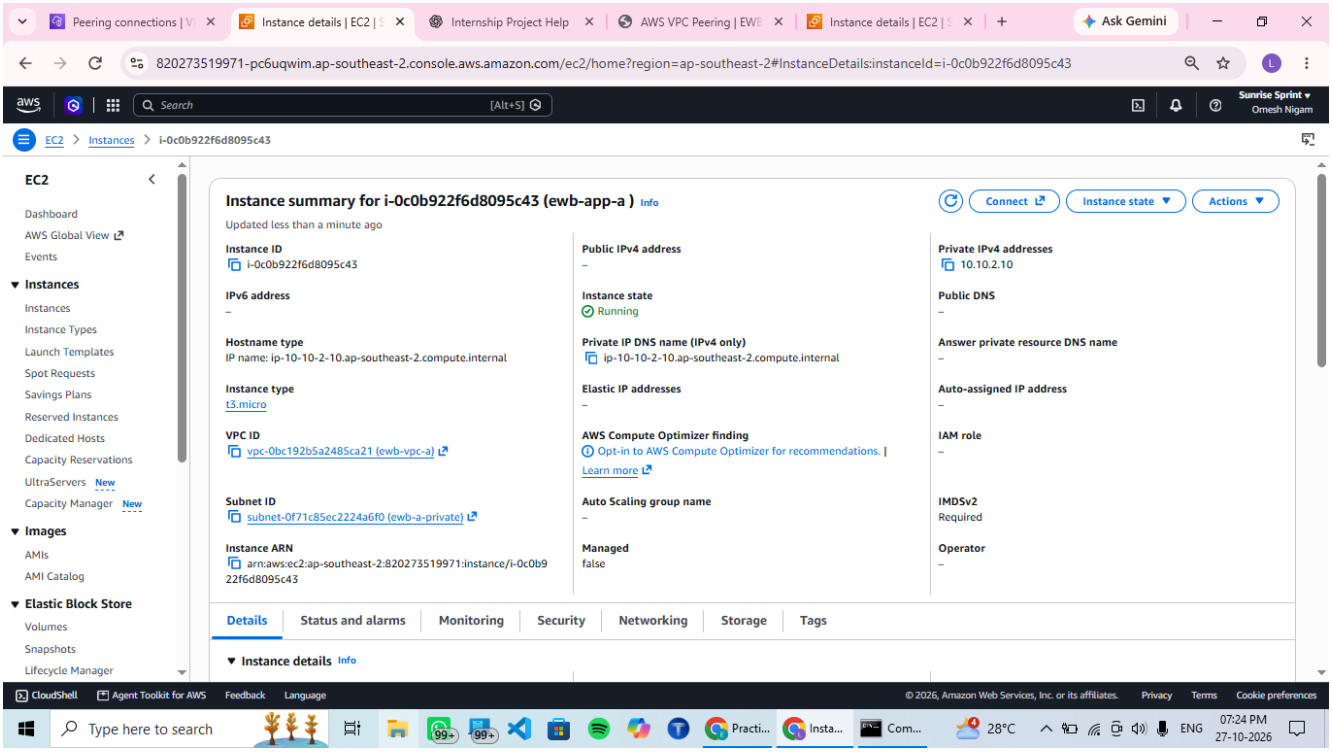
## [SCREENSHOTS GROUP 1: AWS VPC console showing both VPCs and their CIDR blocks]

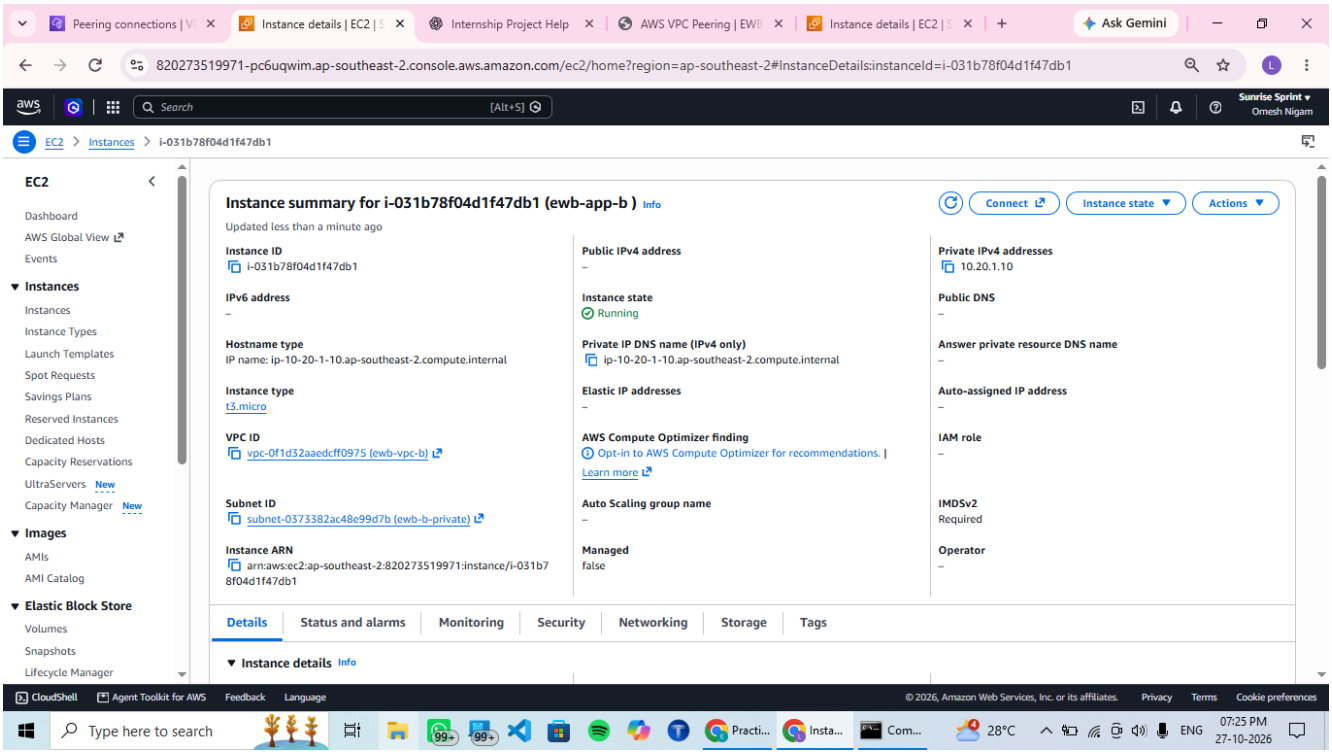


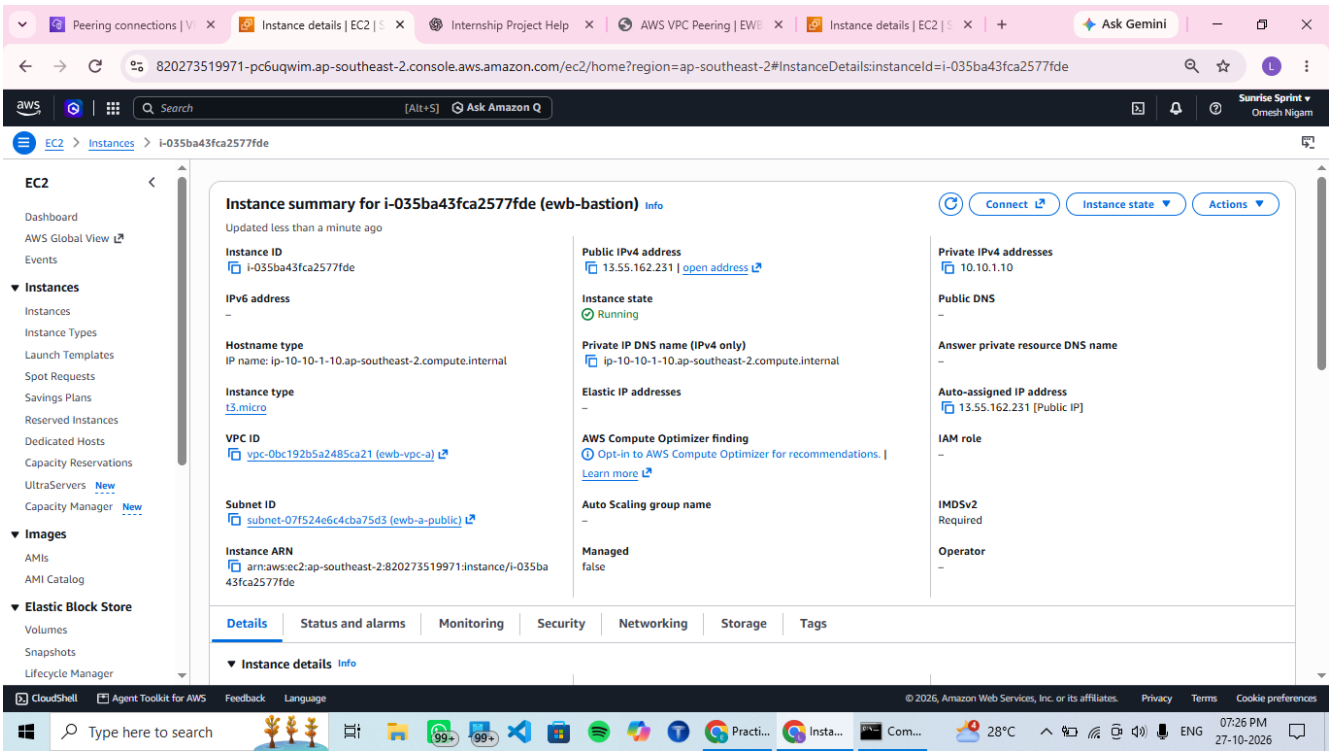



**[SCREENSHOTS GROUP 2: EC2 Instances list/details showing the three instance IDs and private IPs; show App-A and App-B have blank Public IPv4; bastion host configuration]**

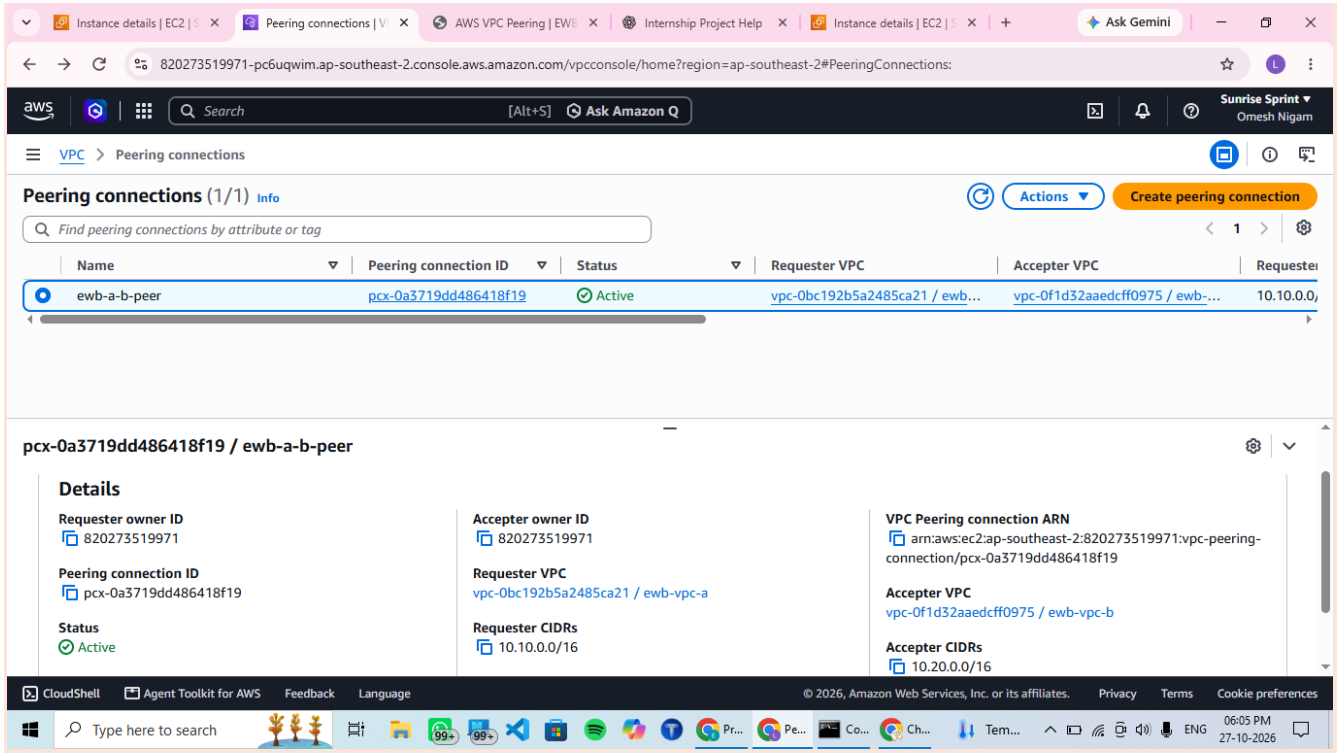


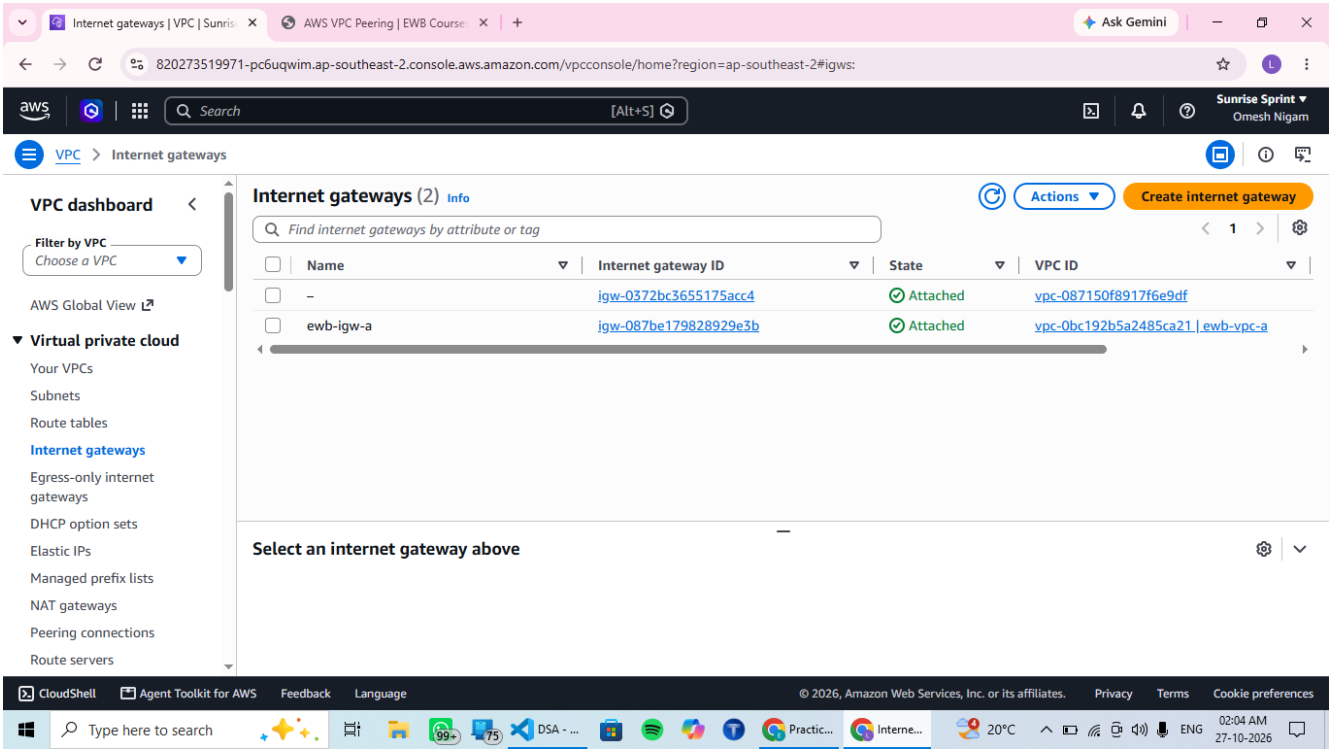

Peering connections | V
Instance details | EC2 |
Internship Project Help
AWS VPC Peering | EWB
Instance details | EC2 |
Ask Gemini
820273519971-pc6uqwim.ap-southeast-2.console.aws.amazon.com/ec2/home?region=ap-southeast-2#InstanceDetails:instanceId=i-031b78f04d1f47db1
Search [Alt+S]
Sunrise Sprint
Omesh Nigam
EC2 > Instances > i-031b78f04d1f47db1
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Instance summary for i-031b78f04d1f47db1 (ewb-app-b ) Info
Updated less than a minute ago
Connect
Instance state
Actions
Instance ID
i-031b78f04d1f47db1
IPv6 address
–
Hostname type
IP name: ip-10-20-1-10.ap-southeast-2.compute.internal
Instance type
t3.micro
VPC ID
vpc-0f1d32aaedcff0975 (ewb-vpc-b)
Subnet ID
subnet-0373382ac48e99d7b (ewb-b-private)
Instance ARN
arn:aws:ec2:ap-southeast-2:820273519971:instance/i-031b78f04d1f47db1
Public IPv4 address
–
Instance state
Running
Private IP DNS name (IPv4 only)
ip-10-20-1-10.ap-southeast-2.compute.internal
Elastic IP addresses
–
AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations.
Learn more
Auto Scaling group name
–
Managed
false
Private IPv4 addresses
10.20.1.10
Public DNS
–
Answer private resource DNS name
–
Auto-assigned IP address
–
IAM role
–
IMDSv2
Required
Operator
–
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance details Info
CloudShell
Agent Toolkit for AWS
Feedback
Language
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
Type here to search
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27-10-2026


Peering connections | V
Instance details | EC2 |
Internship Project Help
AWS VPC Peering | EWB
Instance details | EC2 |
Ask Gemini
820273519971-pc6uqwim.ap-southeast-2.console.aws.amazon.com/ec2/home?region=ap-southeast-2#InstanceDetails:instanceId=i-035ba43fca2577fde
Search [Alt+S] Ask Amazon Q
Sunrise Sprint
Omesh Nigam
EC2 > Instances > i-035ba43fca2577fde
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Instance summary for i-035ba43fca2577fde (ewb-bastion) Info
Updated less than a minute ago
Connect
Instance state
Actions
Instance ID
i-035ba43fca2577fde
IPv6 address
–
Hostname type
IP name: ip-10-10-1-10.ap-southeast-2.compute.internal
Instance type
t3.micro
VPC ID
vpc-0bc192b5a2485ca21 (ewb-vpc-a)
Subnet ID
subnet-07f524e6c4cba75d3 (ewb-a-public)
Instance ARN
arn:aws:ec2:ap-southeast-2:820273519971:instance/i-035ba43fca2577fde
Public IPv4 address
13.55.162.231 | open address
Instance state
Running
Private IP DNS name (IPv4 only)
ip-10-10-1-10.ap-southeast-2.compute.internal
Elastic IP addresses
–
AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations.
Learn more
Auto Scaling group name
–
Managed
false
Private IPv4 addresses
10.10.1.10
Public DNS
–
Answer private resource DNS name
–
Auto-assigned IP address
13.55.162.231 [Public IP]
IAM role
–
IMDSv2
Required
Operator
–
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance details Info
CloudShell
Agent Toolkit for AWS
Feedback
Language
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
Type here to search
28°C
ENG
07:26 PM
27-10-2026

[SCREENSHOT 3: VPC peering connection details showing pcx-0a3719dd486418f19 in Active state]




# 2. Internet Gateway



## 3. Subnets, Route Tables and Associations

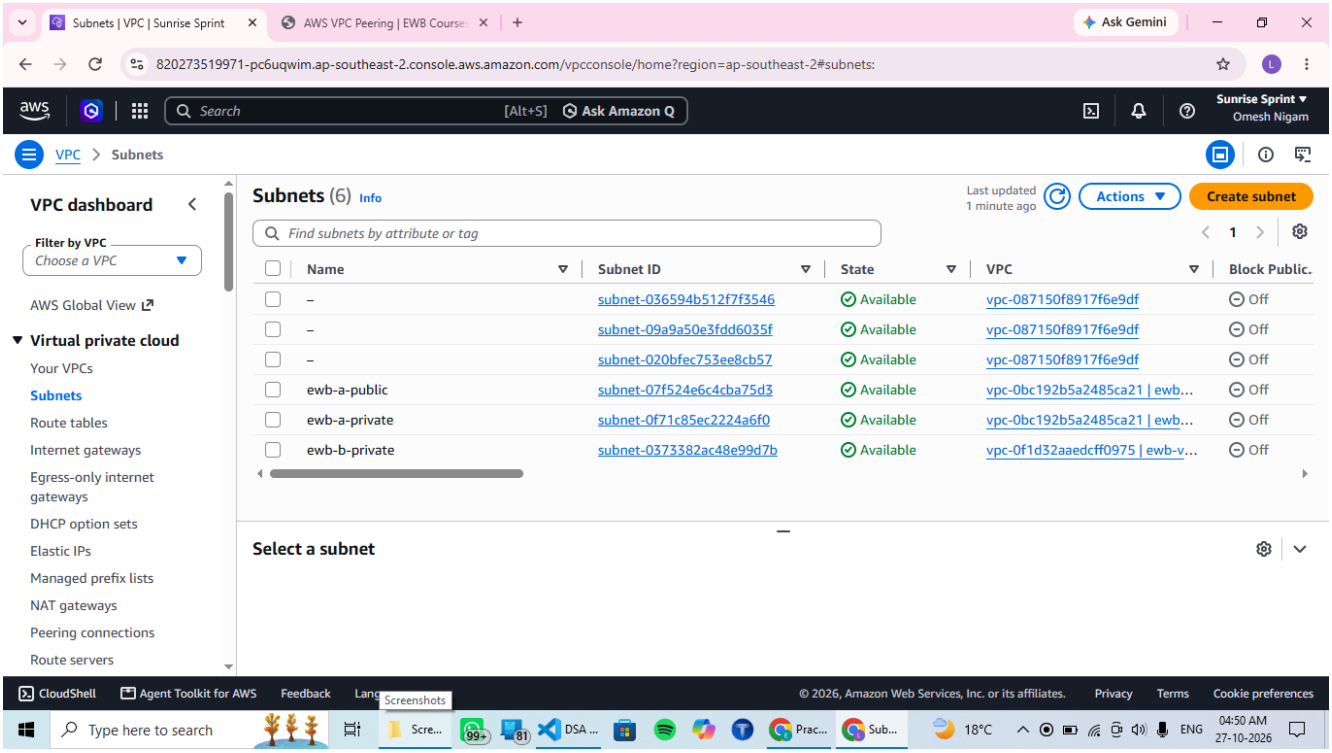
The route tables below keep the AWS-created local route and include the routes needed for peering. The subnet association matters because each subnet uses its associated route table.

| Route table | Subnet association | Required routes |
|---|---|---|
| ewb-rt-a-public<br>rtb-05792380029ea8deb | ewb-a-public (10.10.1.0/24) | 10.10.0.0/16 → local<br>0.0.0.0/0 → igw-087be179828929e3b<br>10.20.0.0/16 → pcx-0a3719dd486418f19 |
| ewb-rt-a-private<br>rtb-0f68fa57f7bec1c58 | ewb-a-private (10.10.2.0/24) | 10.10.0.0/16 → local<br>10.20.0.0/16 → pcx-0a3719dd486418f19 |
| ewb-rt-b-private<br>rtb-08d72c89a6eb81167 | ewb-b-private (10.20.1.0/24) | 10.20.0.0/16 → local<br>10.10.0.0/16 → pcx-0a3719dd486418f19 |

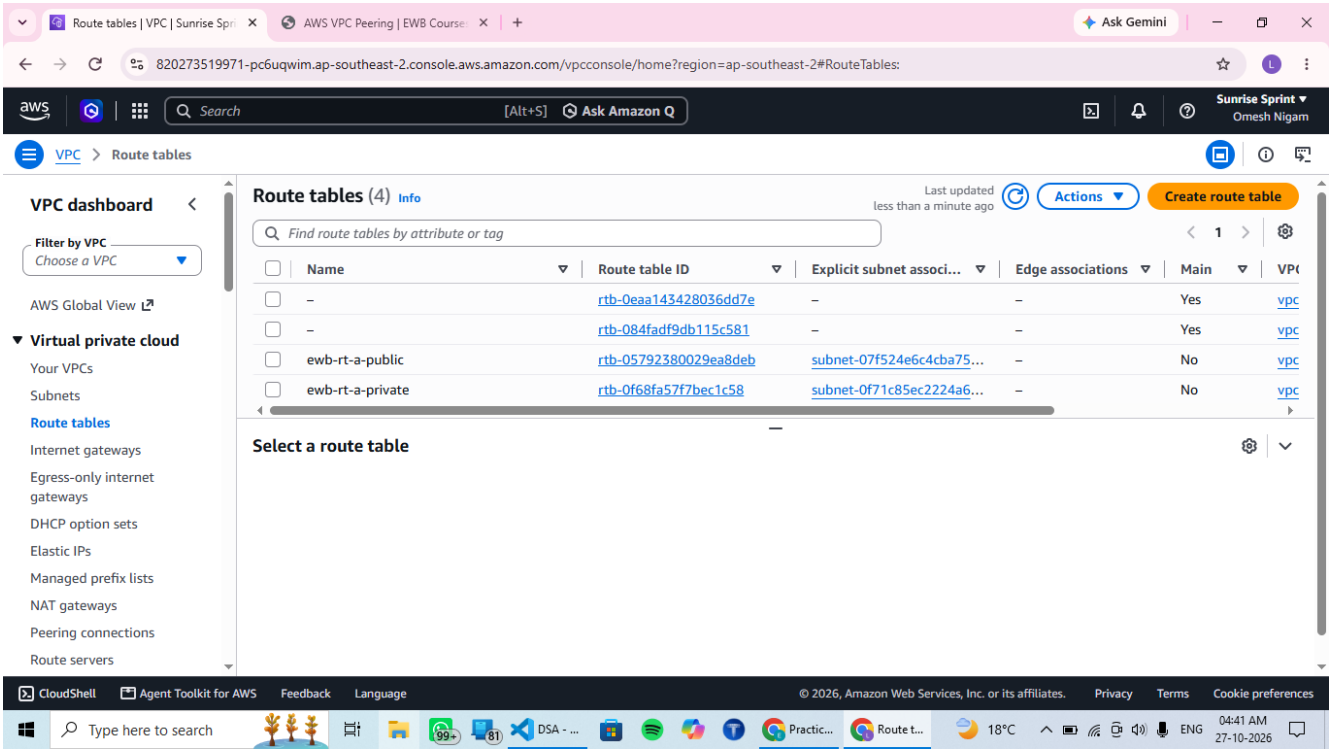
**Forward path:** App-A sends traffic for 10.20.1.10 to the VPC A private route table, where 10.20.0.0/16 points to the peering connection. The public route table also has the peering route so the bastion can reach App-B.

**Return path:** VPC B's private route table sends traffic destined for 10.10.0.0/16 back through the same peering connection. Without the return route, requests may reach the destination but replies cannot return correctly.

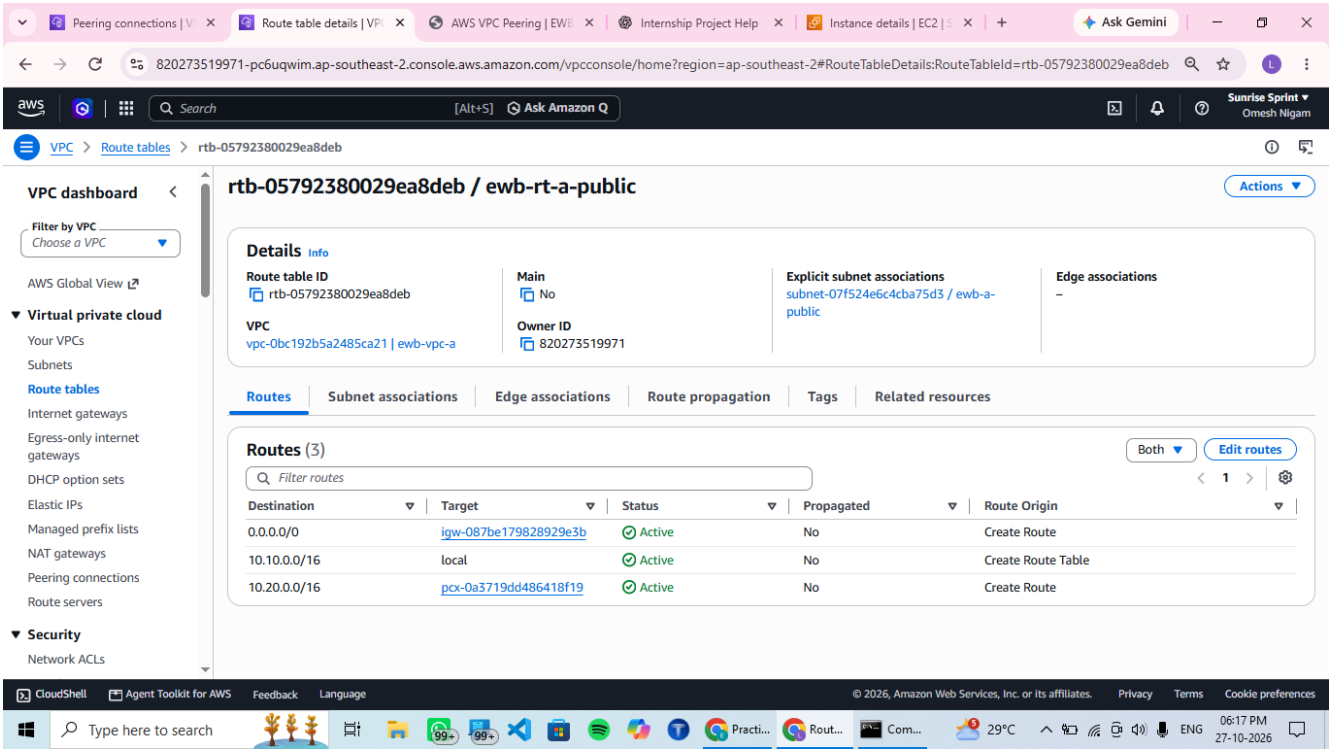
**[SCREENSHOT 4: All Subnets in One Shot]**

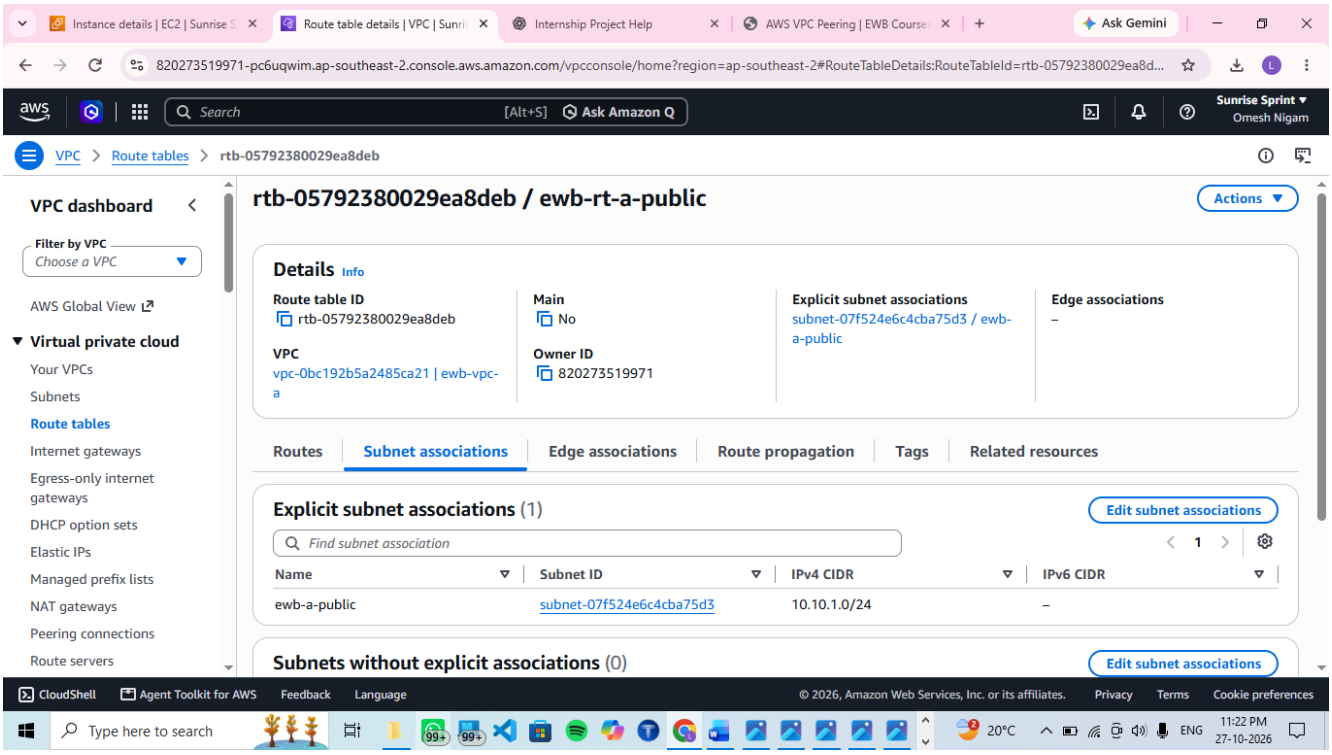


## [SCREENSHOT 5: All Route Tables in one shot]

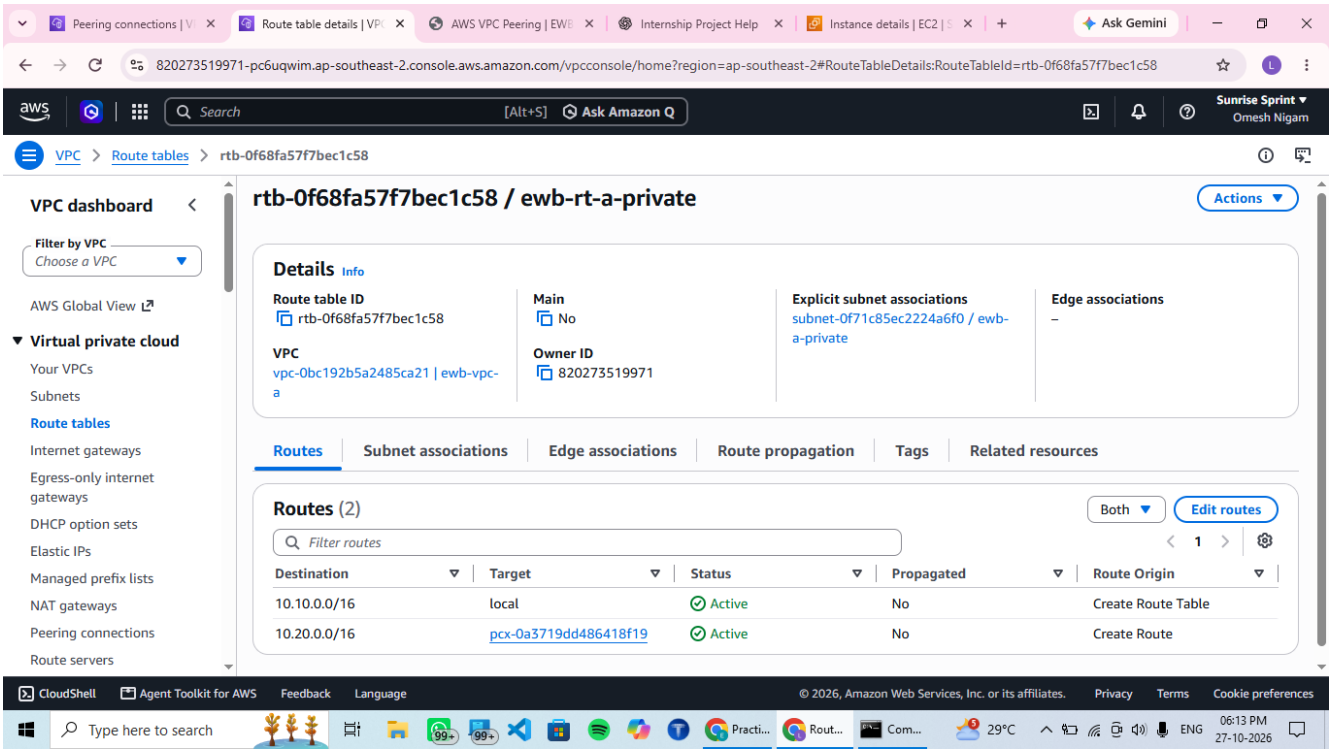


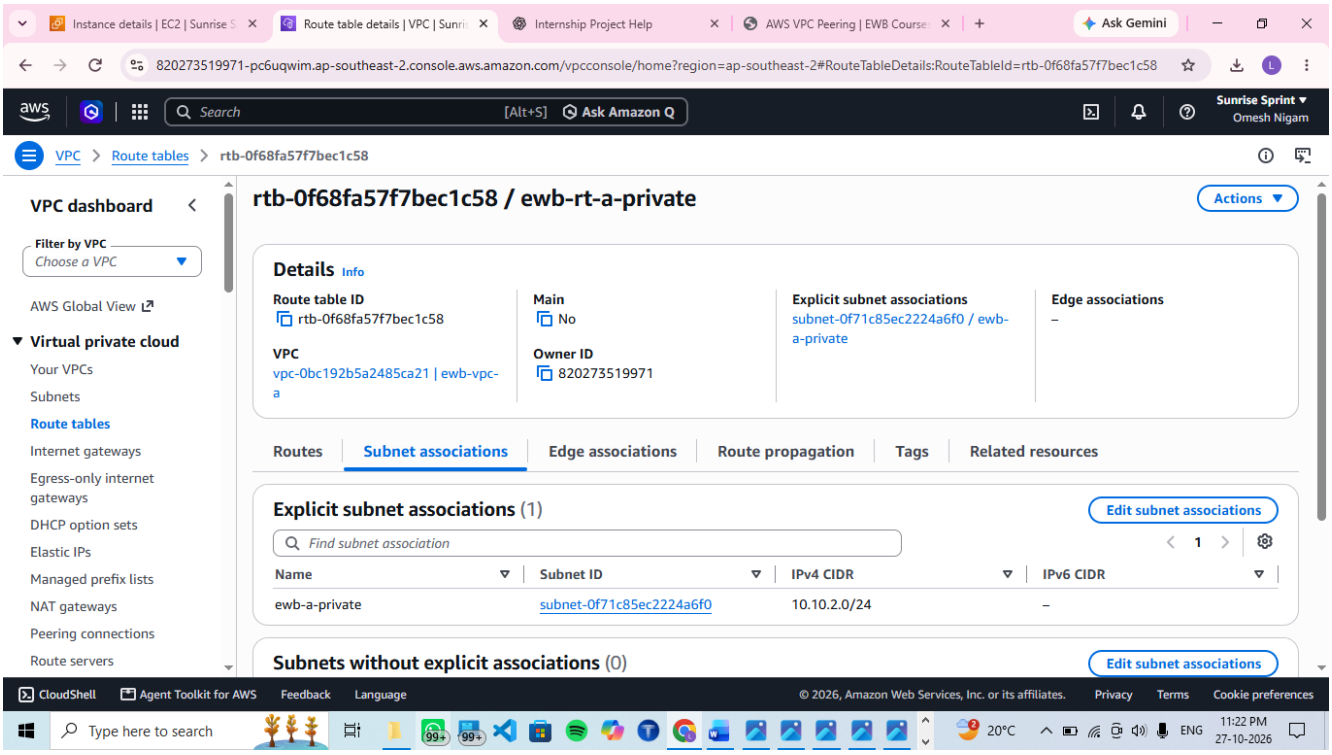

## [SCREENSHOTS GROUP 3: ewb-rt-a-public route table showing routes and associated ewb-a-public subnet]



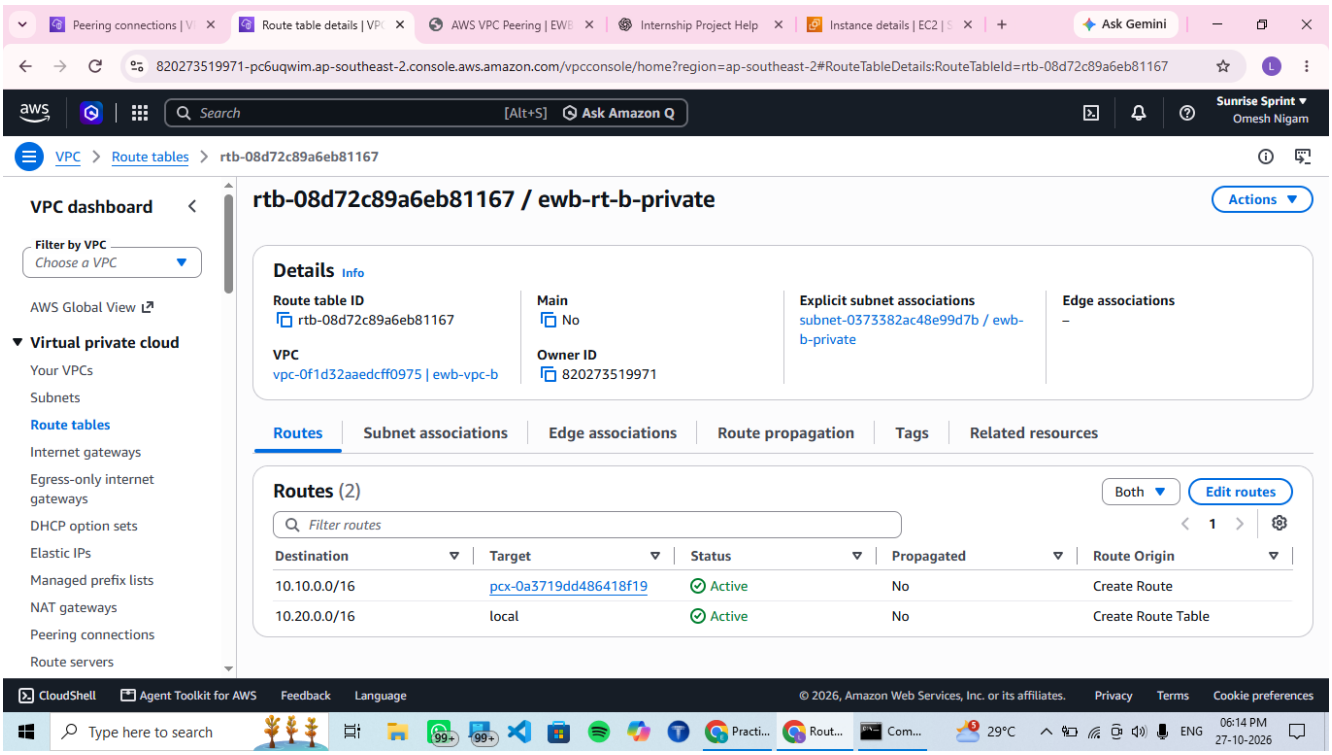


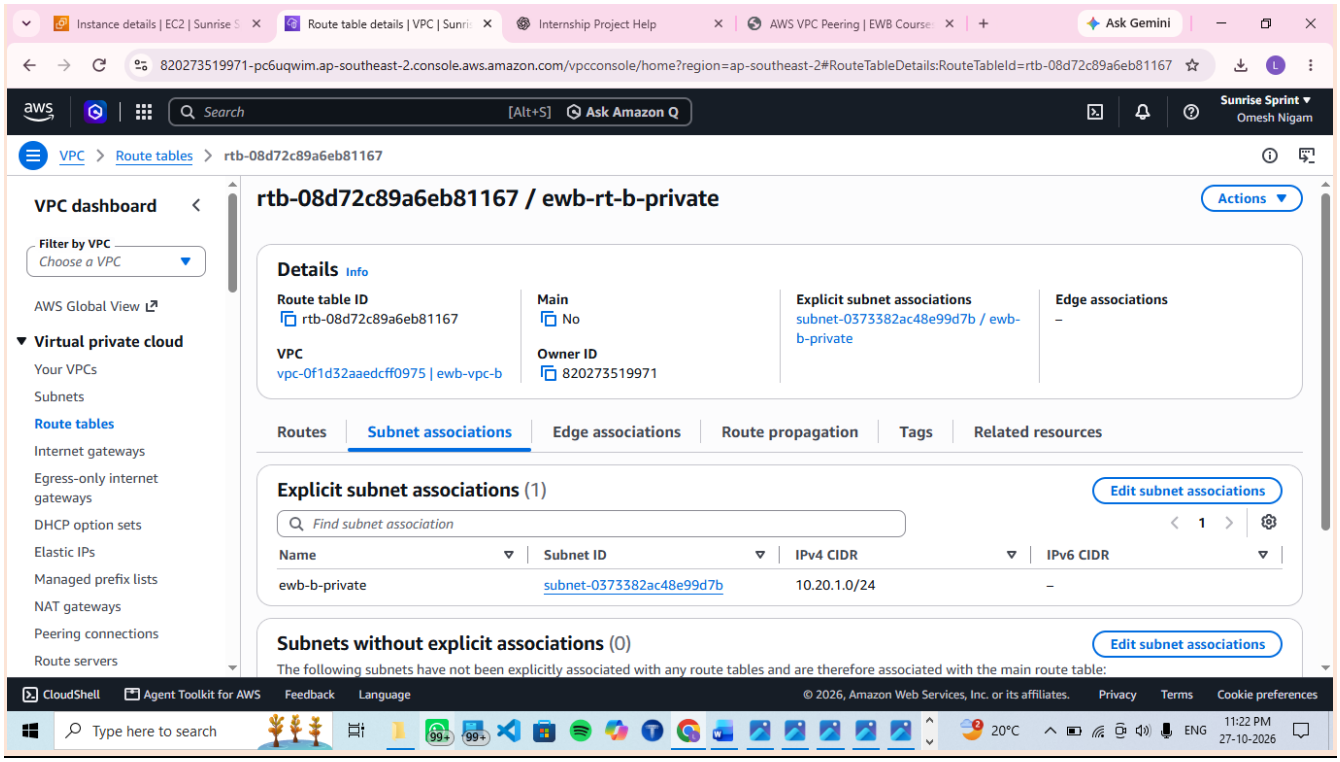

## [SCREENSHOTS GROUP 4: ewb-rt-a-private route table showing peering route and associated ewb-a-private subnet]






## [SCREENSHOTS GROUP 5: ewb-rt-b-private route table showing return peering route and associated ewb-b-private subnet]



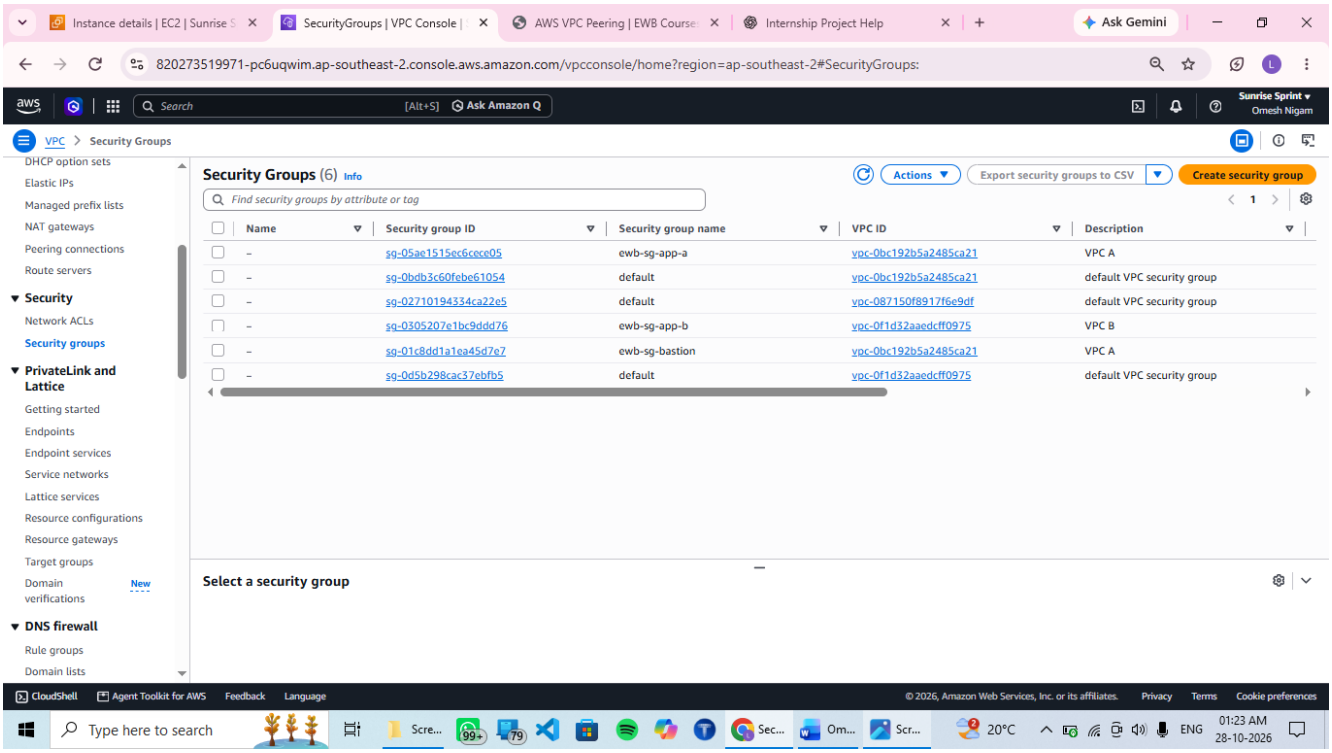



## 4. Security Groups

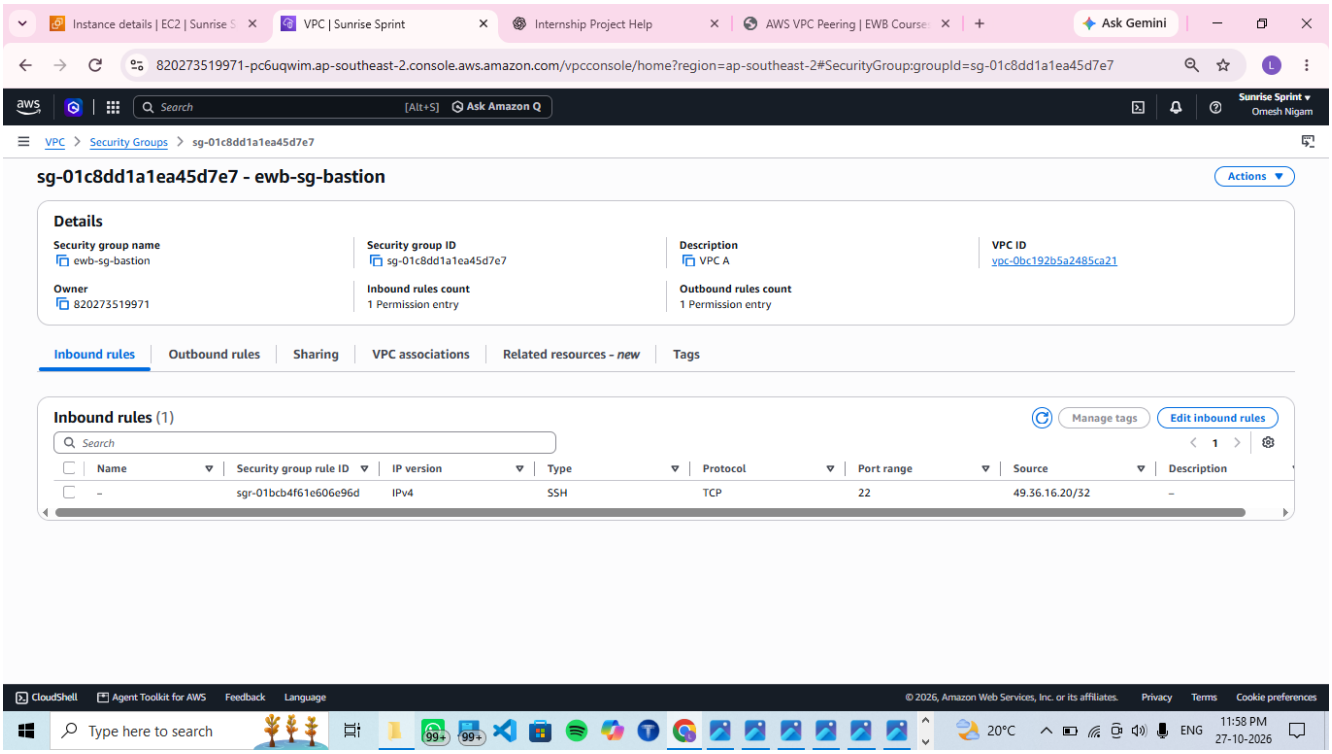
| Security group | Inbound rules to document |
|---|---|
| ewb-sg-bastion | SSH / TCP 22 from your current public IP address (/32). |
| ewb-sg-app-a | SSH / TCP 22 from 10.10.1.10/32 (bastion private IP); ICMP Echo Request from 10.20.1.10/32. |
| ewb-sg-app-b | SSH / TCP 22 from 10.10.1.10/32 (bastion private IP); ICMP Echo Request from 10.10.2.10/32. |

Outbound rules: the lab security groups use allow-all outbound traffic (0.0.0.0/0).

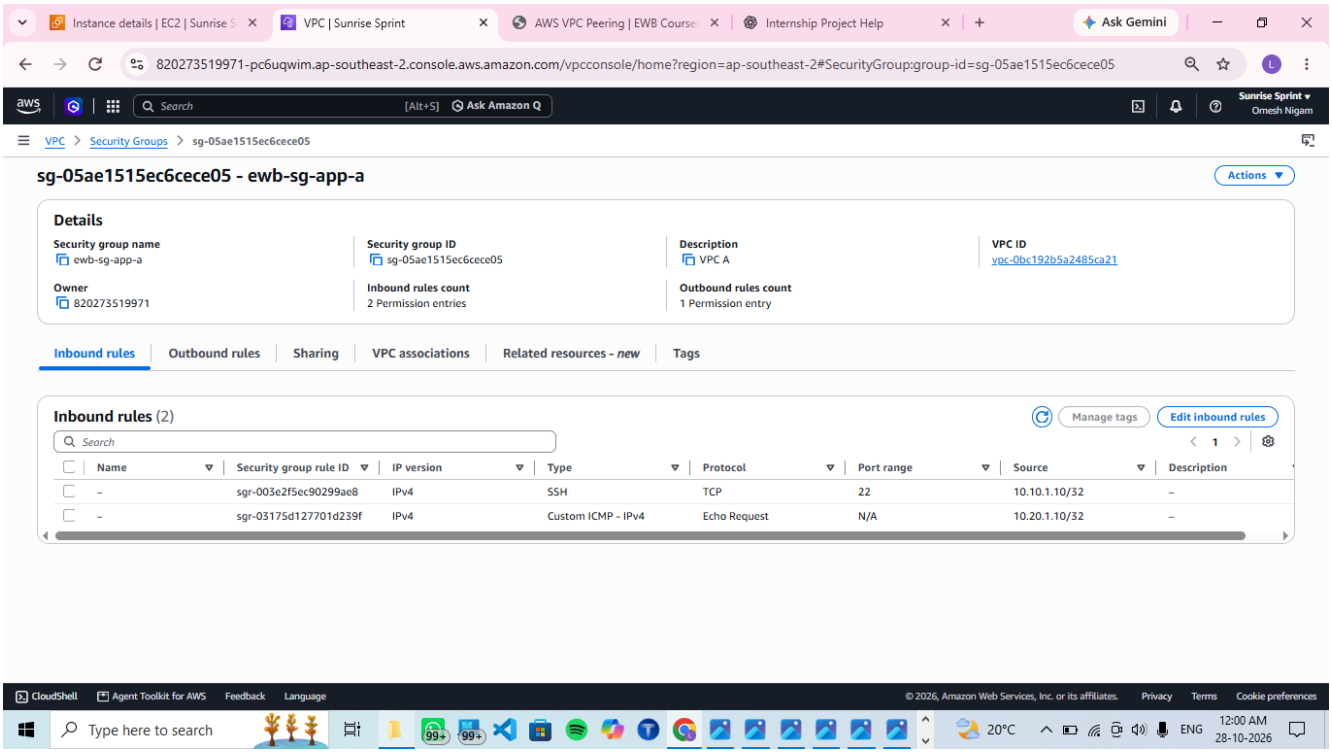
## [SCREENSHOT 6: All Security Groups in one shot]



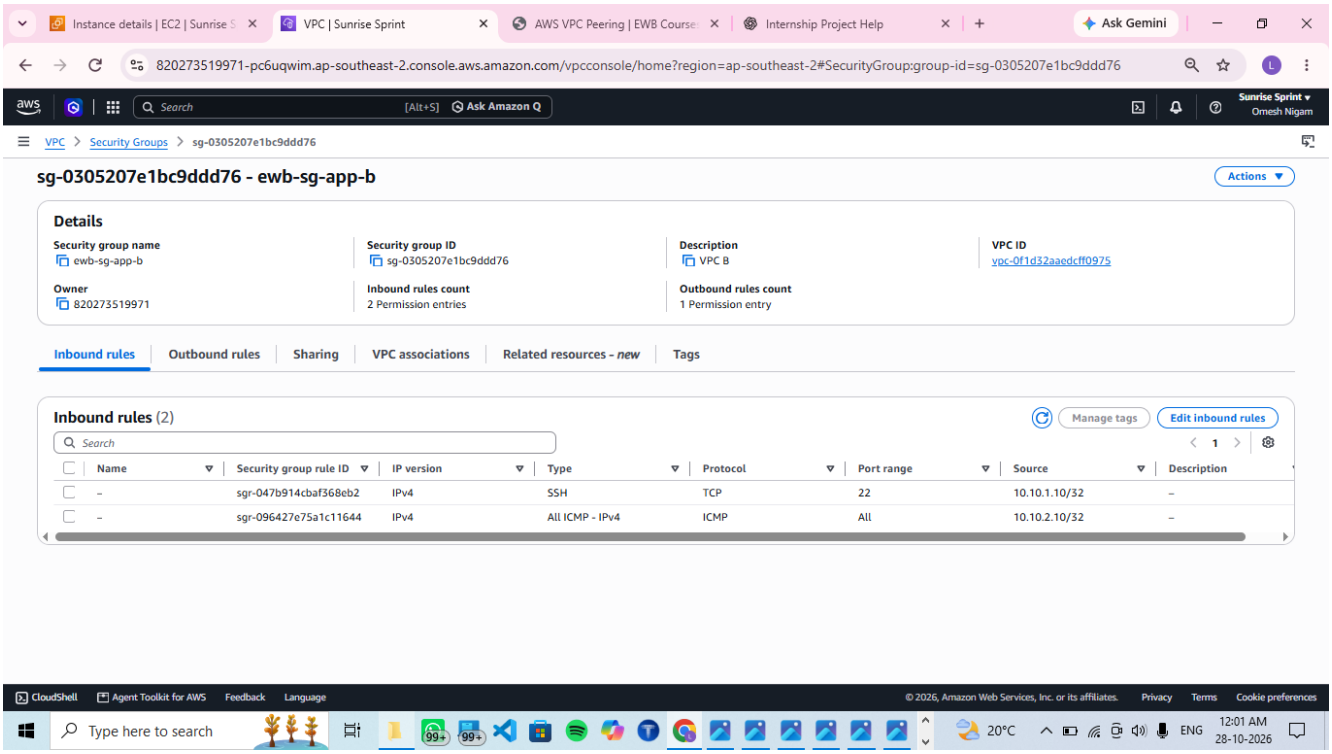

## [SCREENSHOT 7: ewb-sg-bastion inbound rules screenshot]



## [SCREENSHOT 8: ewb-sg-app-a inbound rules screenshot]




## [SCREENSHOT 9: ewb-sg-app-b inbound rules screenshot, with SSH and ICMP rules visible]



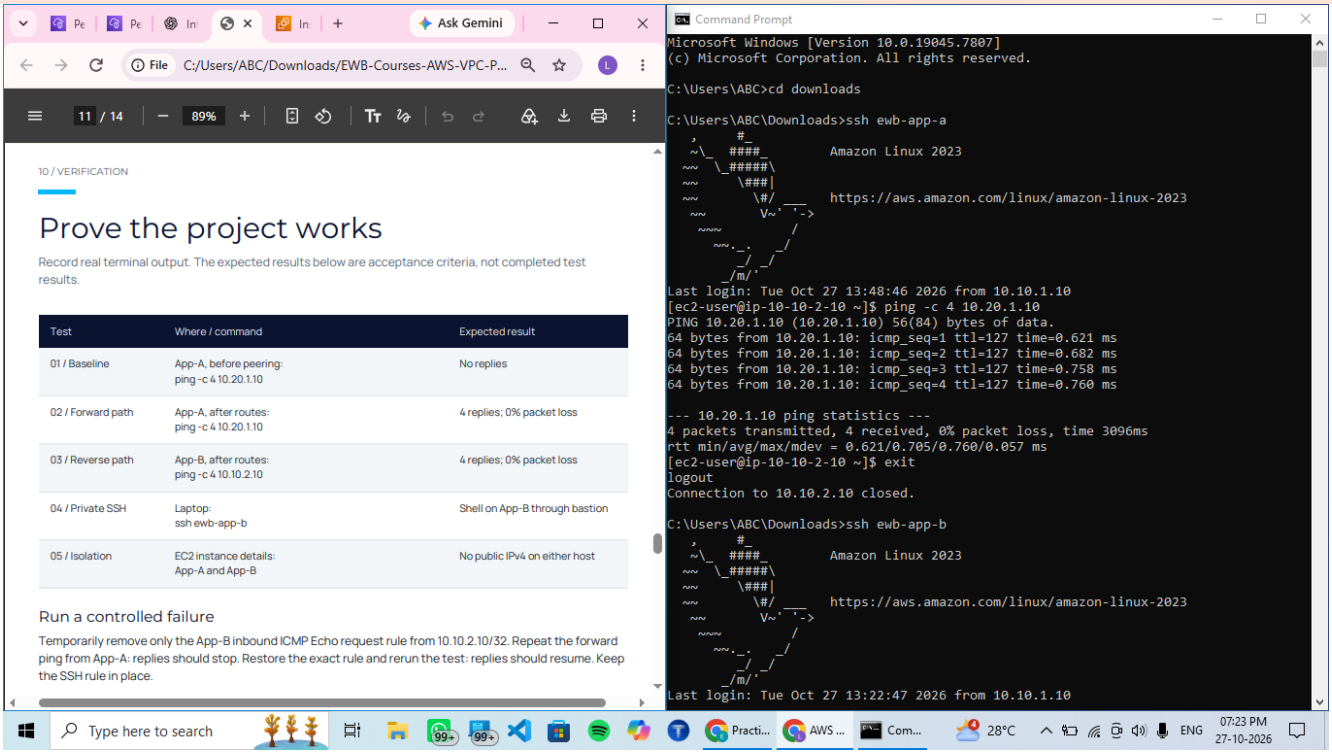
# 5. Connectivity Verification

## 5.1 Forward path — App-A to App-B

Command executed on App-A: ping -c 4 10.20.1.10

Result: [Confirm from your captured terminal output; expected result is 4 replies and 0% packet loss.]

**[SCREENSHOT 10: Terminal on App-A showing hostname / hostname -I and ping -c 4 10.20.1.10 with final packet summary]**

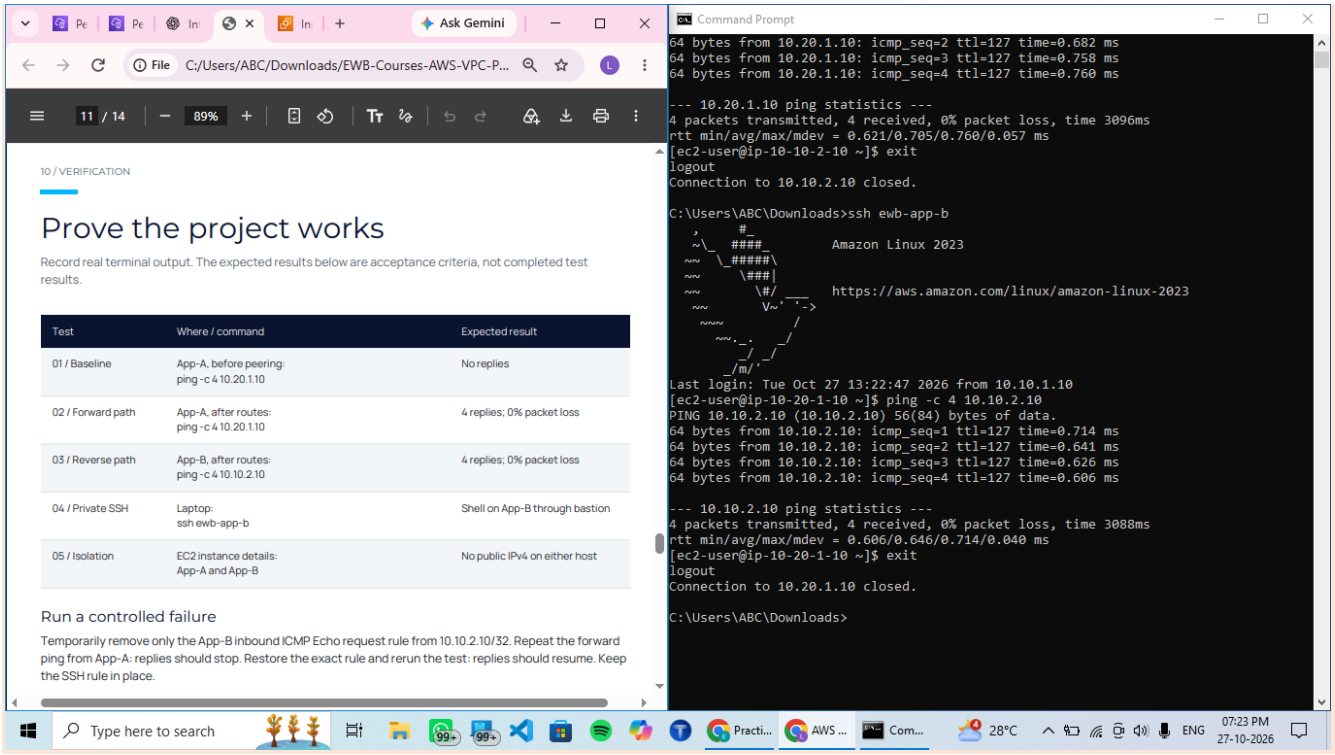



## 5.2 Reverse path — App-B to App-A

Command executed on App-B: ping -c 4 10.10.2.10

Result: [Confirm from your captured terminal output; expected result is 4 replies and 0% packet loss.]

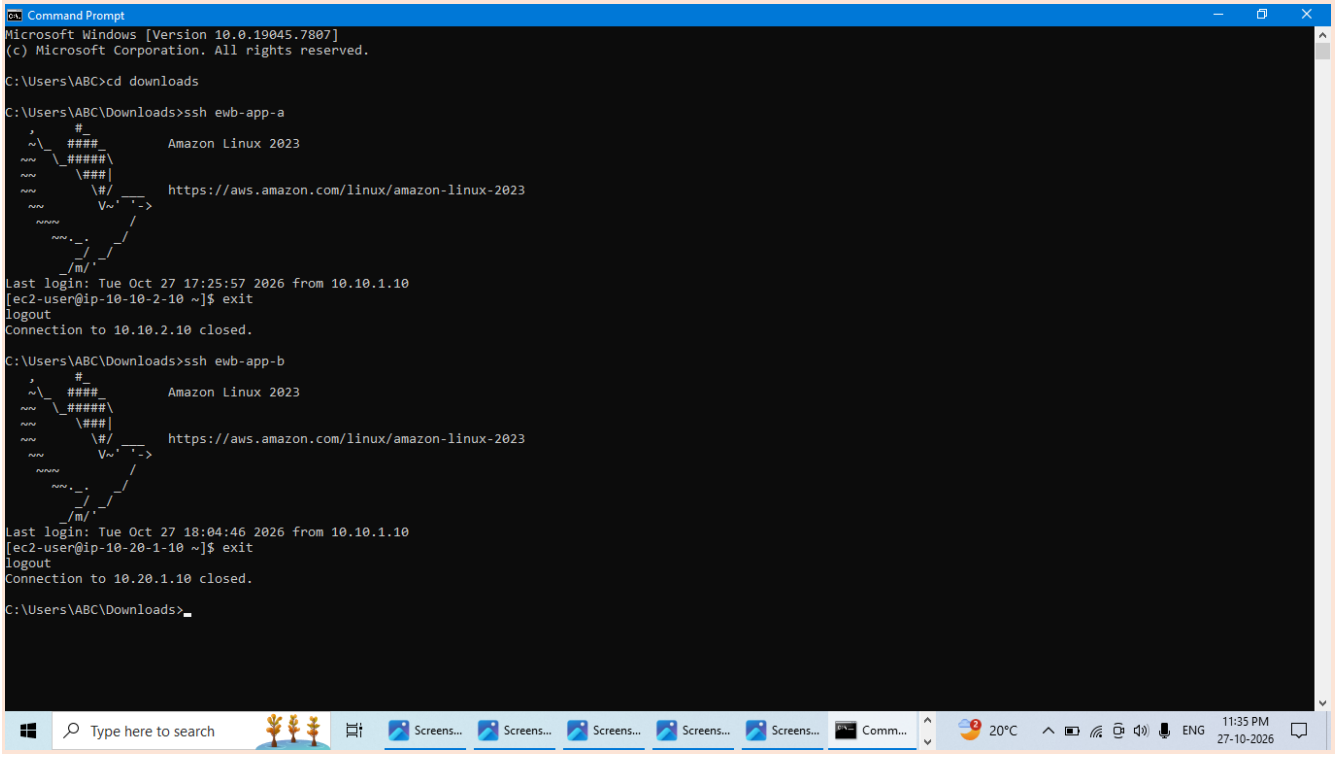
**[SCREENSHOT 11: Terminal on App-B showing hostname / hostname -I and ping -c 4 10.10.2.10 with final packet summary]**




## 5.3 Private SSH through the bastion

From the laptop, the command `ssh ewb-app-b` uses SSH ProxyJump through the bastion to reach App-B's private IP. The private instances do not need public IPv4 addresses for this connection.

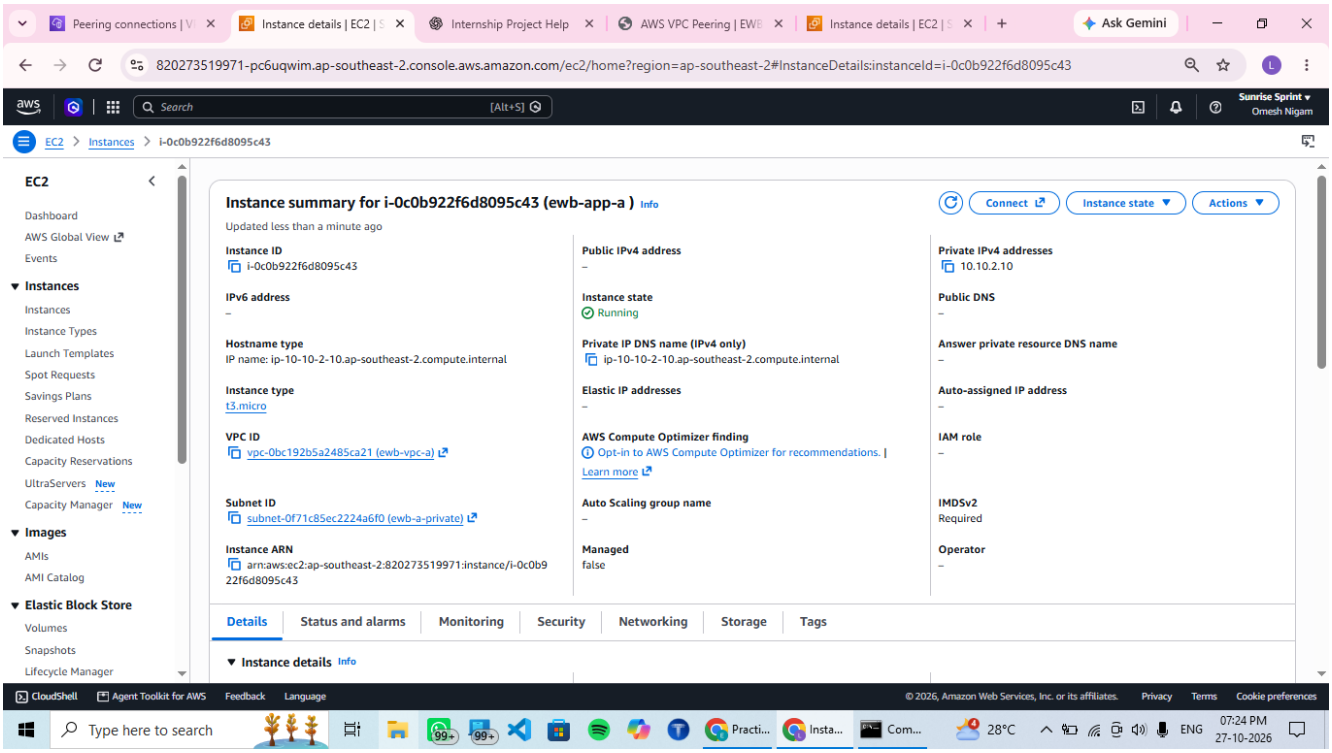
**[SCREENSHOT 12: Laptop terminal showing successful `ssh ewb-app-b` login and App-B shell/hostname]**

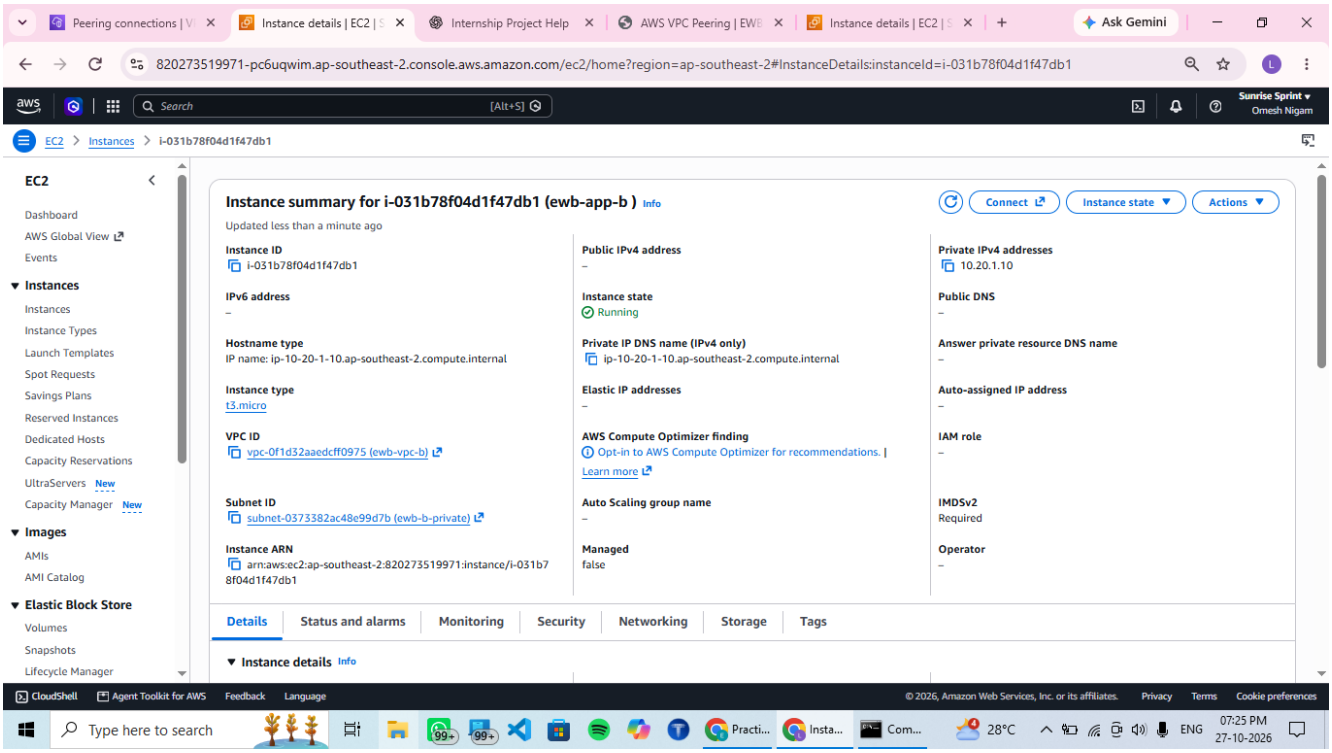


## 5.4 Isolation — no public IPv4 on private hosts

App-A (10.10.2.10) and App-B (10.20.1.10) are private instances and should have no Public IPv4 address.

**[SCREENSHOT 13: EC2 instance details/list showing blank Public IPv4 for both ewb-app-a and ewb-app-b]**



### 5.5 Baseline failure before peering

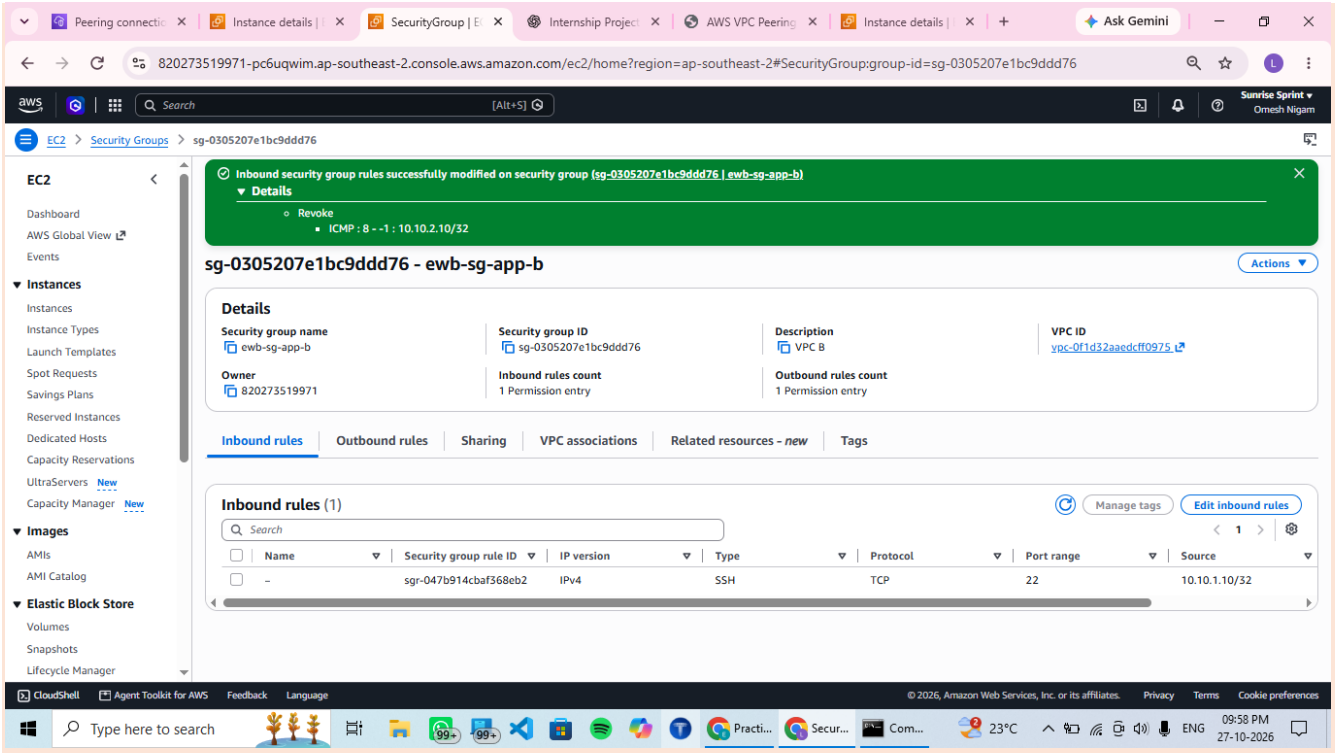
Workbook acceptance criterion: before peering, a ping from App-A to 10.20.1.10 should receive no replies. In this lab, peering and routes had already been configured before the baseline was recorded, so a genuine pre-peering baseline screenshot was not captured. Do not present a later test as if it were a pre-peering result.

## 6. Controlled ICMP Failure and Recovery

A controlled failure was tested by temporarily removing only the App-B inbound ICMP Echo Request rule from source 10.10.2.10/32, while keeping the SSH rule in place. The forward ping from App-A was then repeated. After restoring the exact ICMP rule, the ping was repeated to confirm recovery.

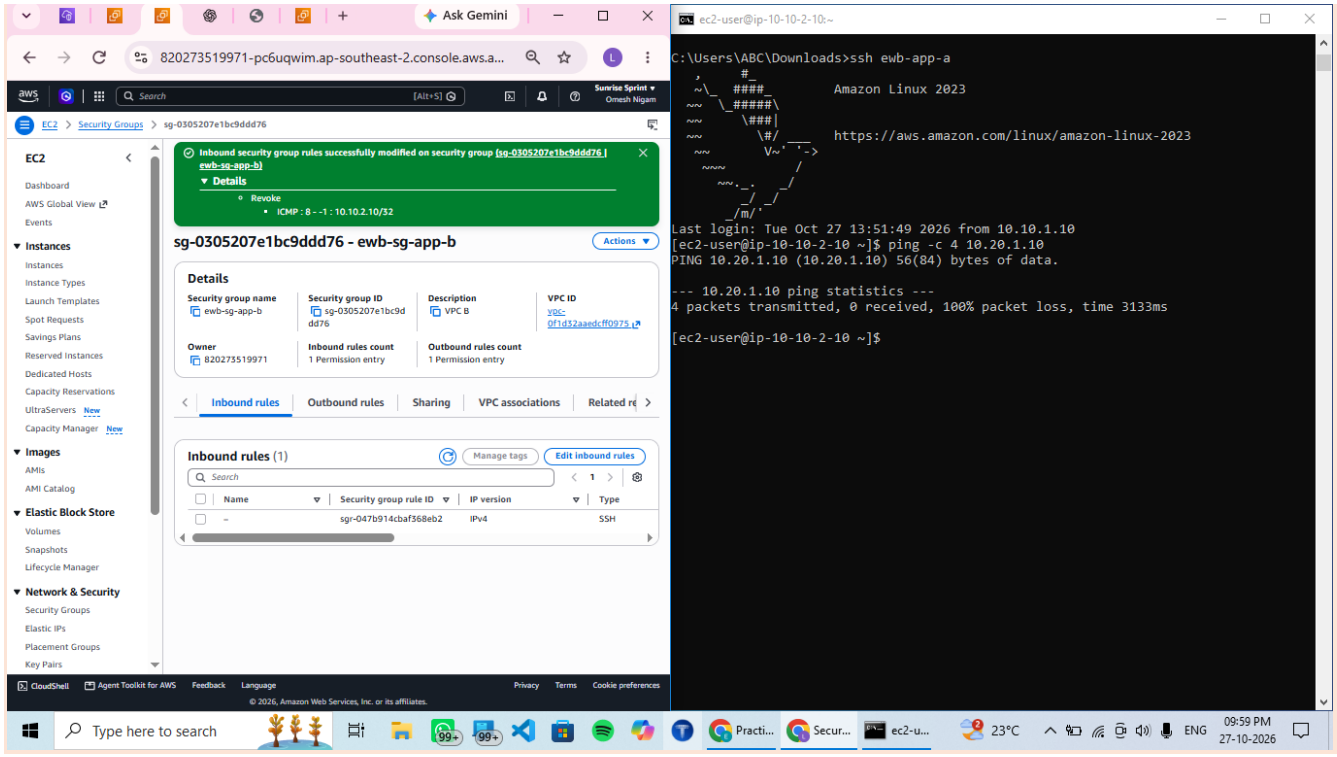
### 6.1 ICMP rule temporarily removed — expected failure

**[SCREENSHOT 14: Screenshot of App-B security group inbound rules with only the ICMP Echo Request rule temporarily absent, OR terminal showing failed forward ping during this period]**



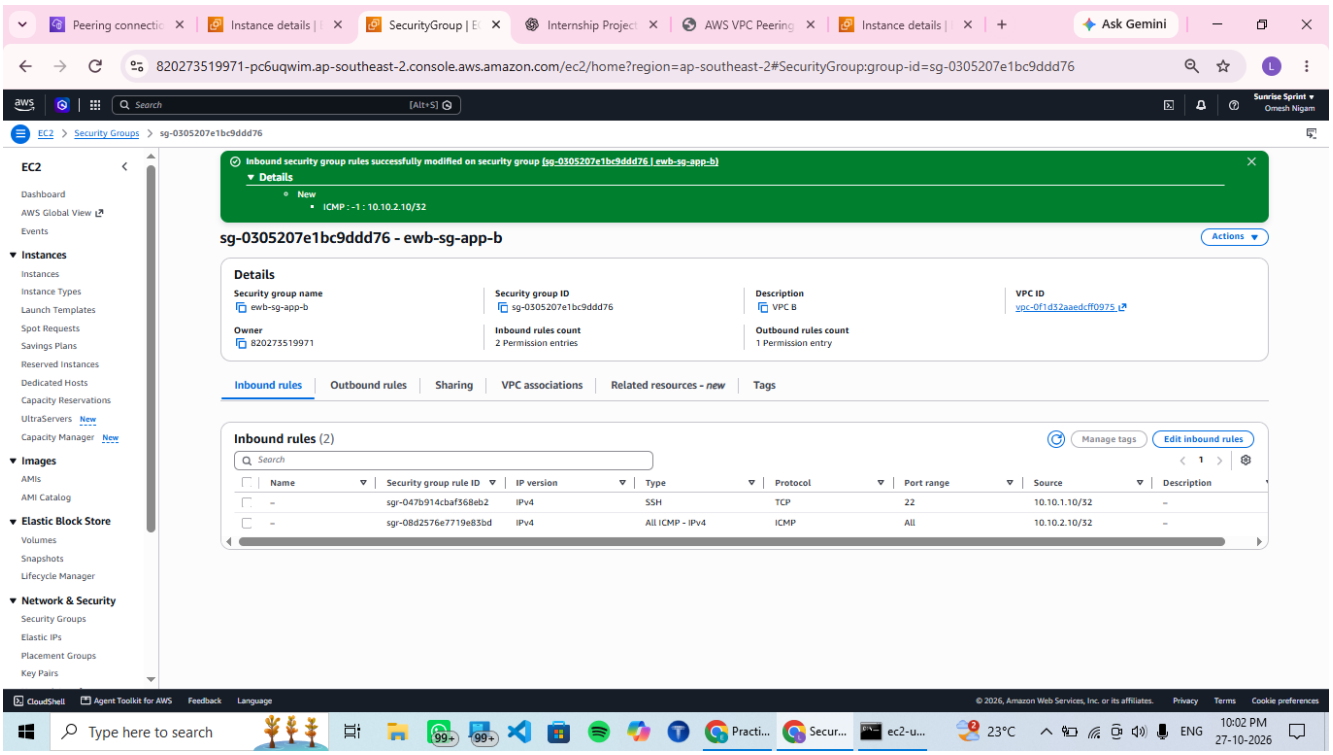

### 6.2 Forward ping failed during controlled test [NEGATIVE TEST]

**[SCREENSHOT 15: App-A terminal showing ping -c 4 10.20.1.10 failing(100% Packet Loss) while the ICMP rule was removed]**

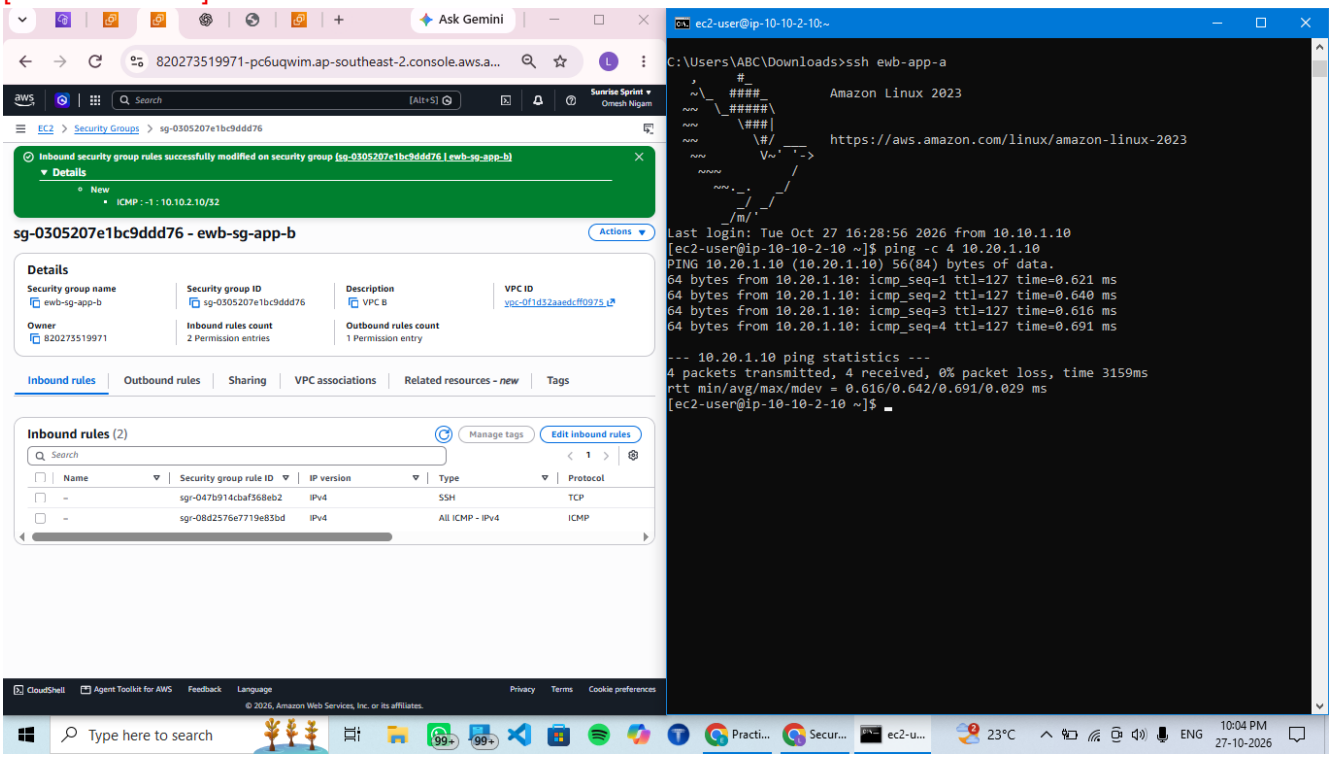



## 6.3 ICMP rule restored — connectivity recovered

**[SCREENSHOT 16: App-B security group showing restored ICMP Echo Request from 10.10.2.10/32]**



**[SCREENSHOT 17: App-A terminal showing successful ping -c 4 10.20.1.10 after restoring the rule]**
[POSTIVE TEST]




**Explanation**: the peering connection and routes provide a network path, but the destination security group can still block ICMP. Removing the Echo Request rule causes the ping to fail; restoring it allows the ping to succeed again.

## 7. Key Learnings

- VPC peering connects the two VPCs, but it does not automatically add routes to route tables.
- Both the forward route and return route are required for two-way communication.
- Security groups must allow the required traffic. An Active peering connection alone does not guarantee a successful ping.
- App-B SSH traffic through ProxyJump comes from the bastion's private IP, 10.10.1.10, so the App-B SSH rule uses 10.10.1.10/32 as its source.
- VPC B cannot use VPC A's Internet Gateway through VPC peering.
- Peering is non-transitive: connecting A–B and A–C does not automatically connect B–C.

## 8. Viva Questions

**1. Why is the return route needed?**

**Ans.** A return route is required because network communication involves traffic travelling in both directions. When App-A sends a request to App-B, App-B must have a route back to App-A's network to send the response. Without the return route, communication may fail even if the forward route exists.

**2. Why is the App-B SSH source the bastion private IP?**

**Ans.** The App-B Security Group allows SSH traffic (TCP port 22) from the bastion's private IP address, 10.10.1.10. This restricts SSH access to the bastion host instead of allowing direct SSH access from any source. It improves security by providing a controlled entry point to the private EC2 instance.

**3. Why does Active peering not guarantee a successful ping?**

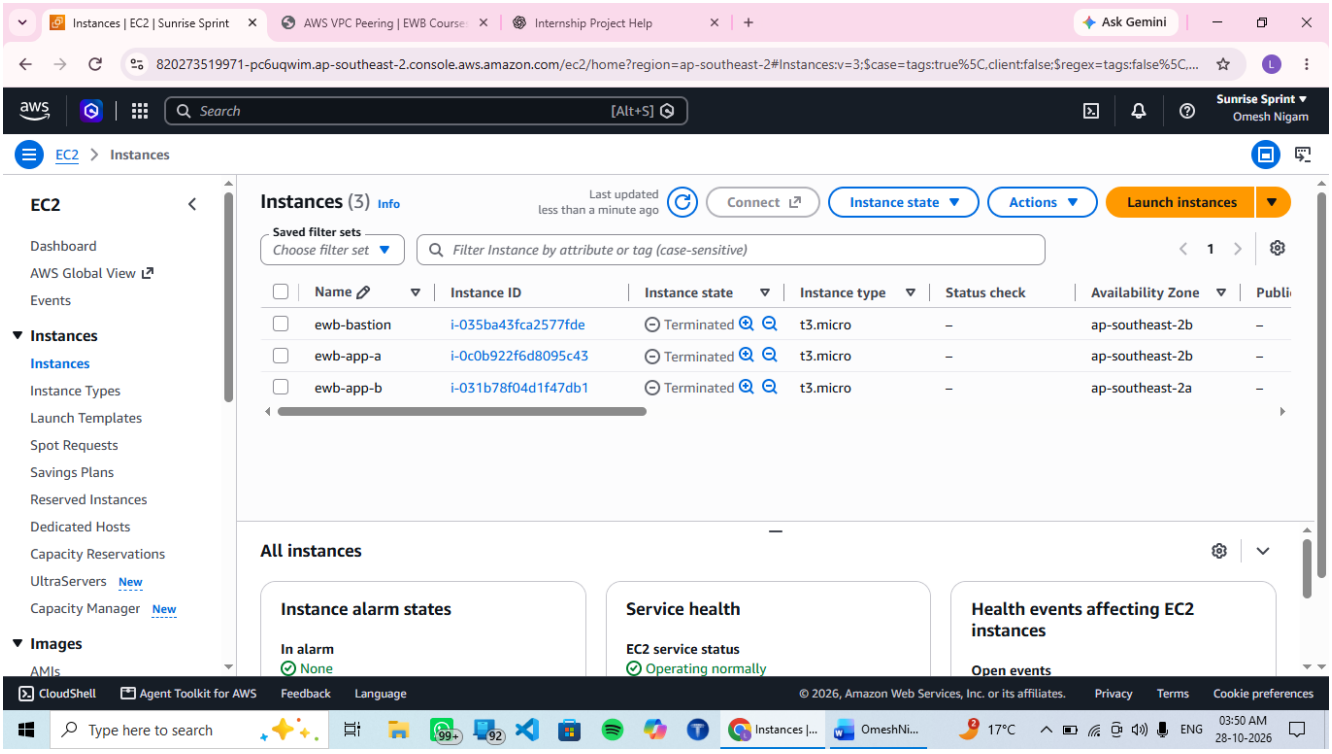
**Ans.** An Active VPC Peering connection indicates that the peering connection has been established. However, successful ping also depends on correct routes in both VPCs, Security Group rules allowing ICMP traffic, and any applicable network ACL rules. If any required configuration is missing or blocking traffic, ping may fail.

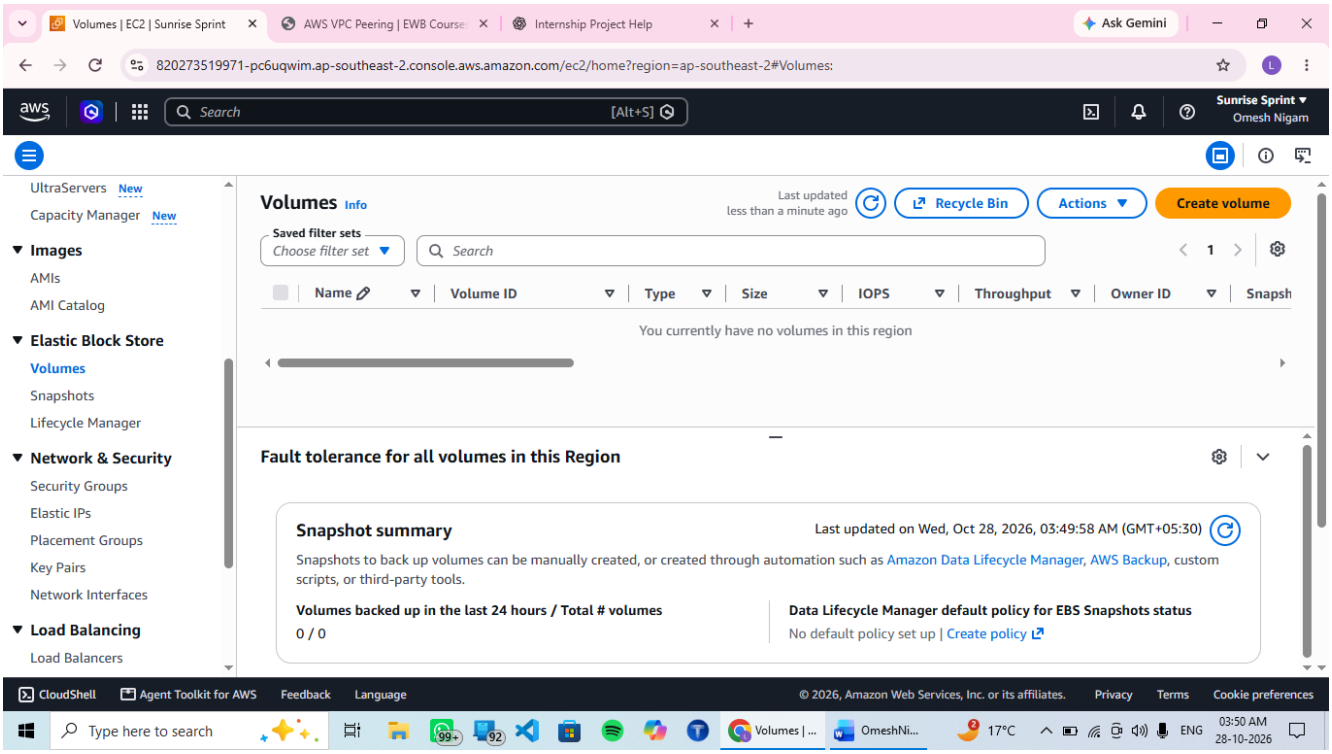
**4. Can VPC B use VPC A's internet gateway over peering?**

**Ans.** No. VPC Peering does not allow VPC B to use VPC A's Internet Gateway to access the internet. VPC Peering supports private communication between the two VPCs. If VPC B requires internet access, it needs its own Internet Gateway for public-subnet resources or a suitable NAT-based solution for private-subnet resources.

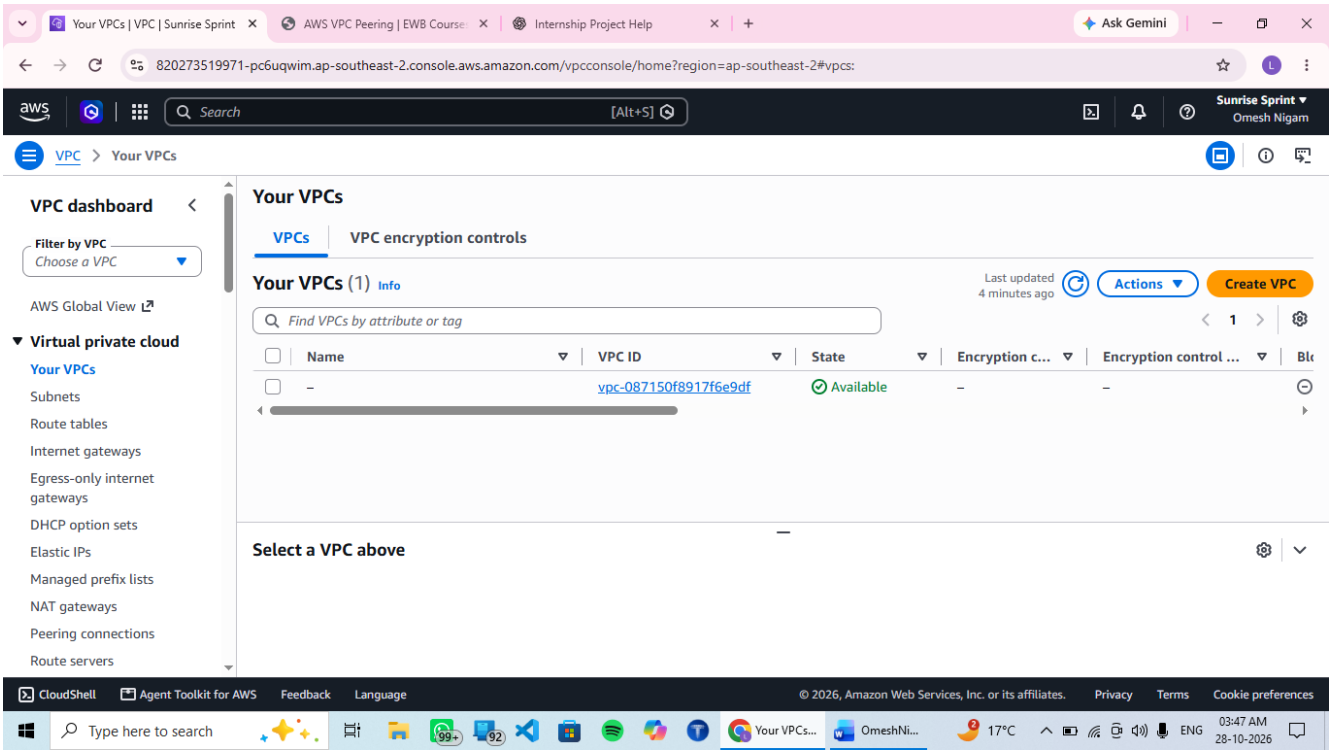
## 9. Cleanup Evidence

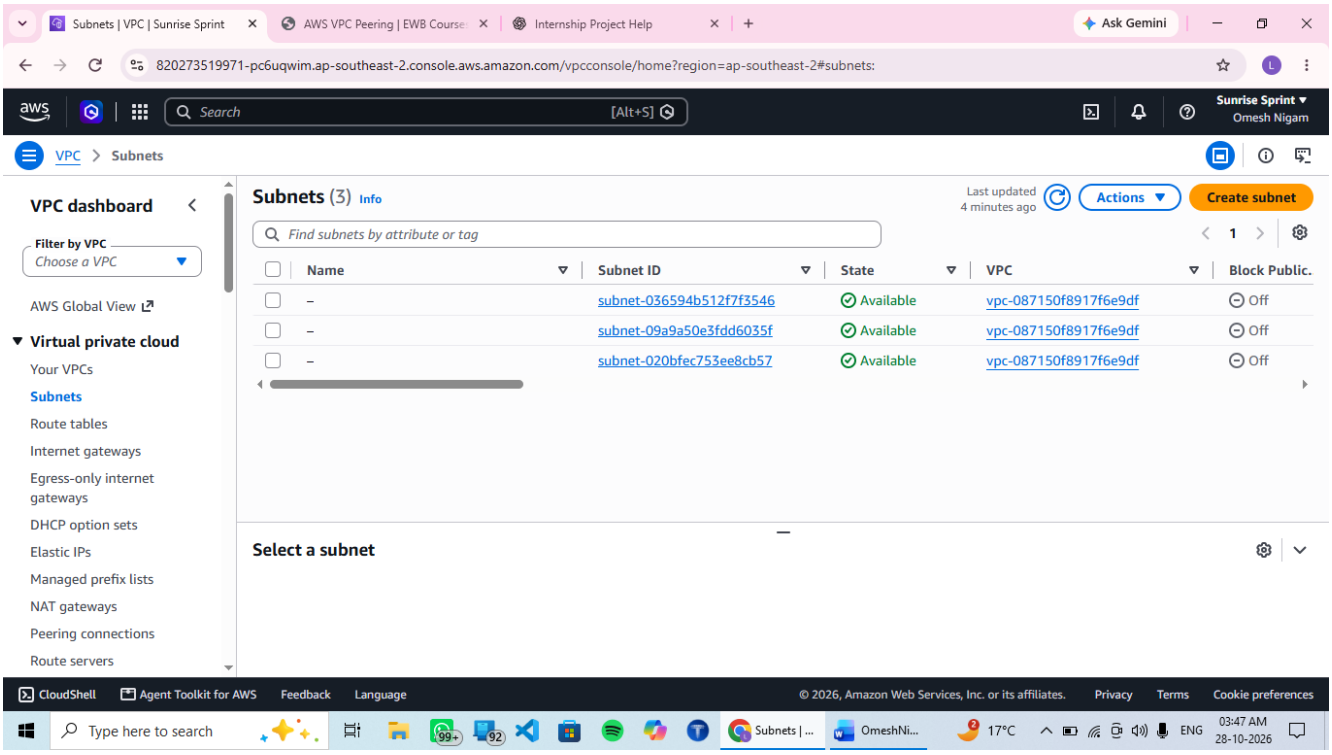
**[SCREENSHOTS GROUP 6: Cleanup evidence: EC2 instances terminated and no leftover lab EBS volumes/snapshots or Elastic IPs]**

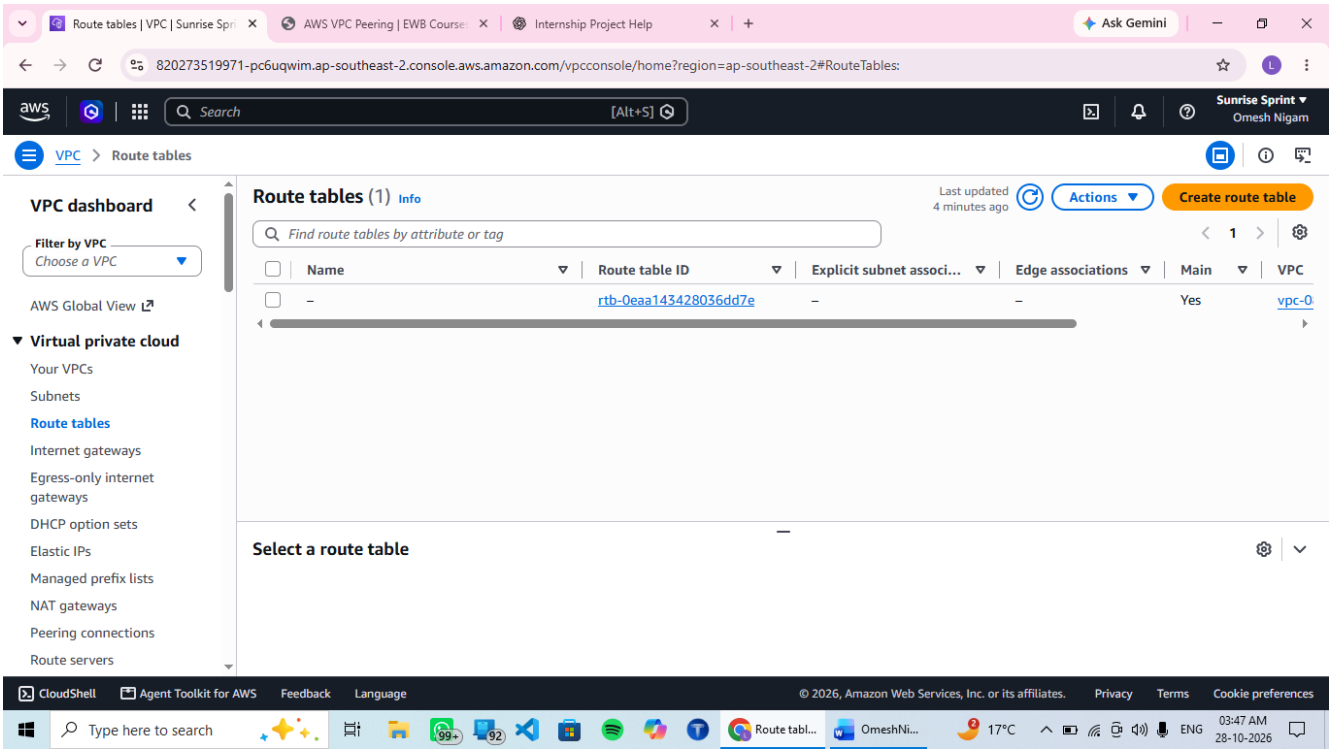


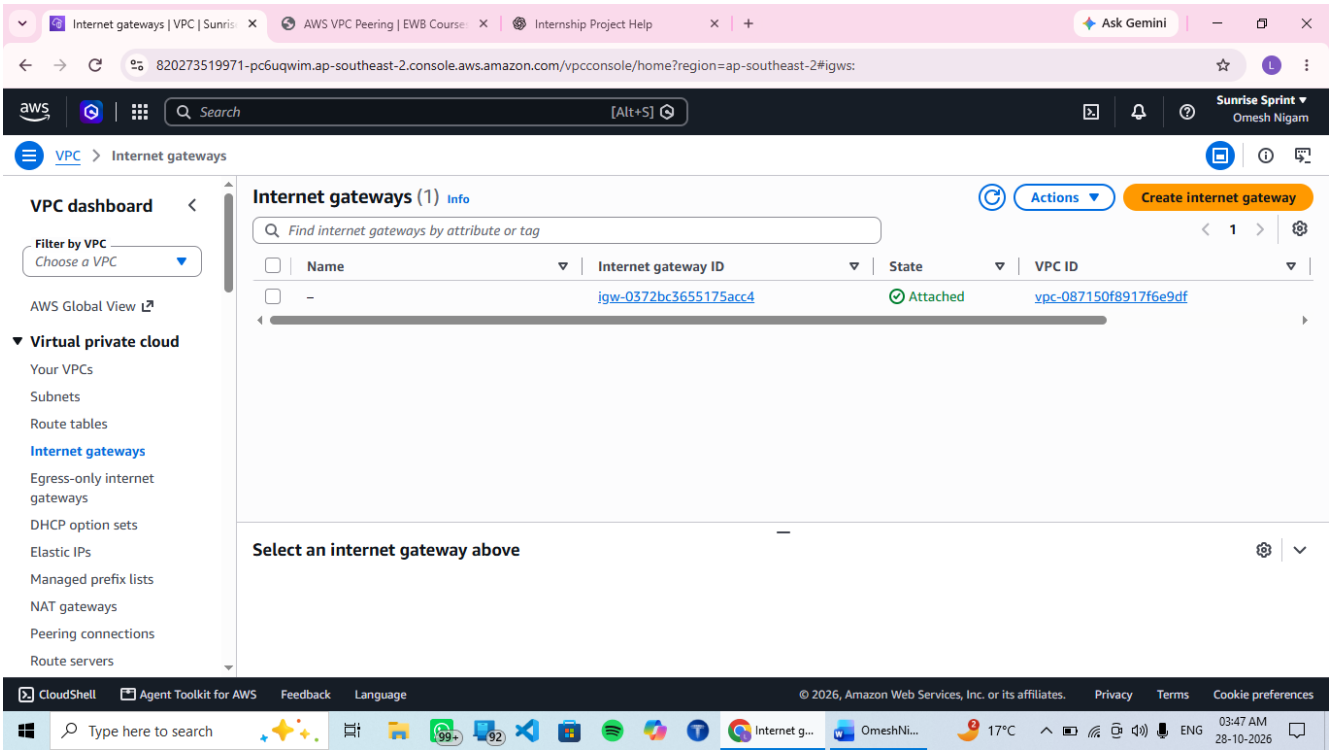


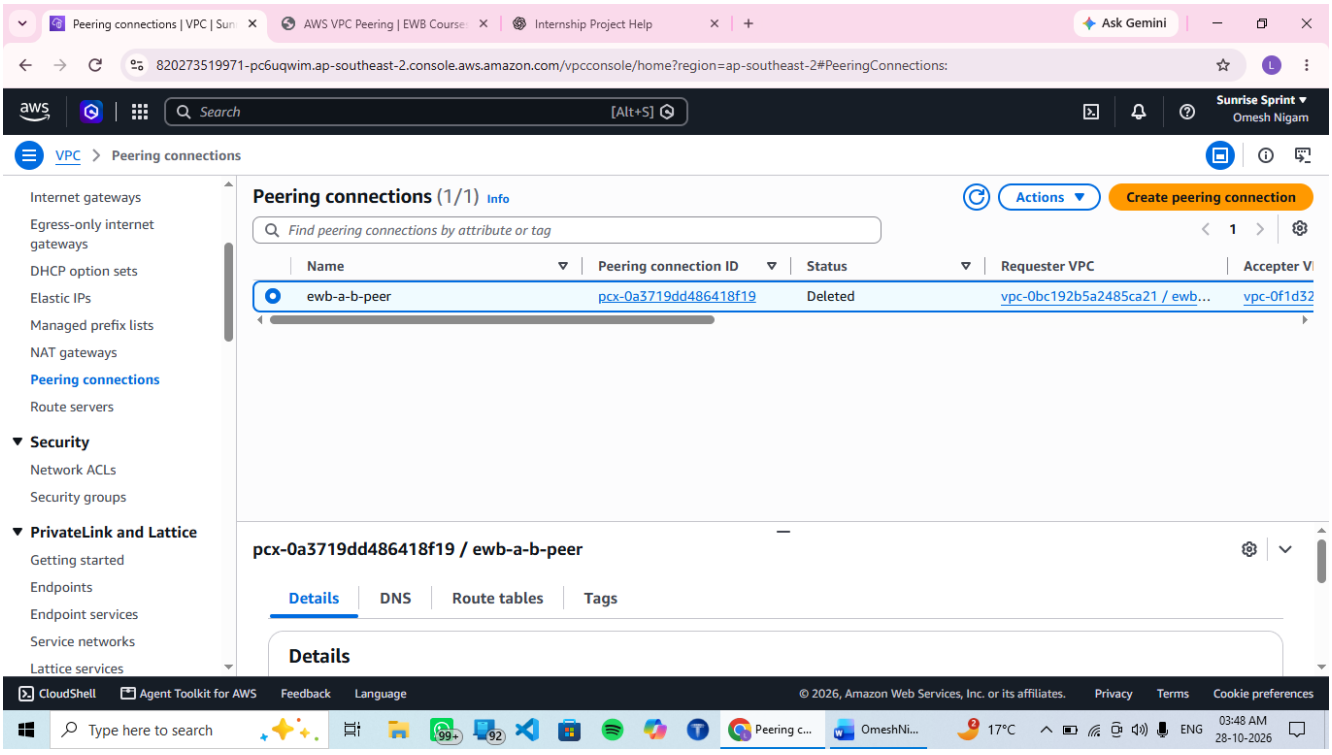

## [SCREENSHOTS GROUP 7: Cleanup evidence: VPC console showing lab peering, subnets, route tables, security groups and VPCs removed]

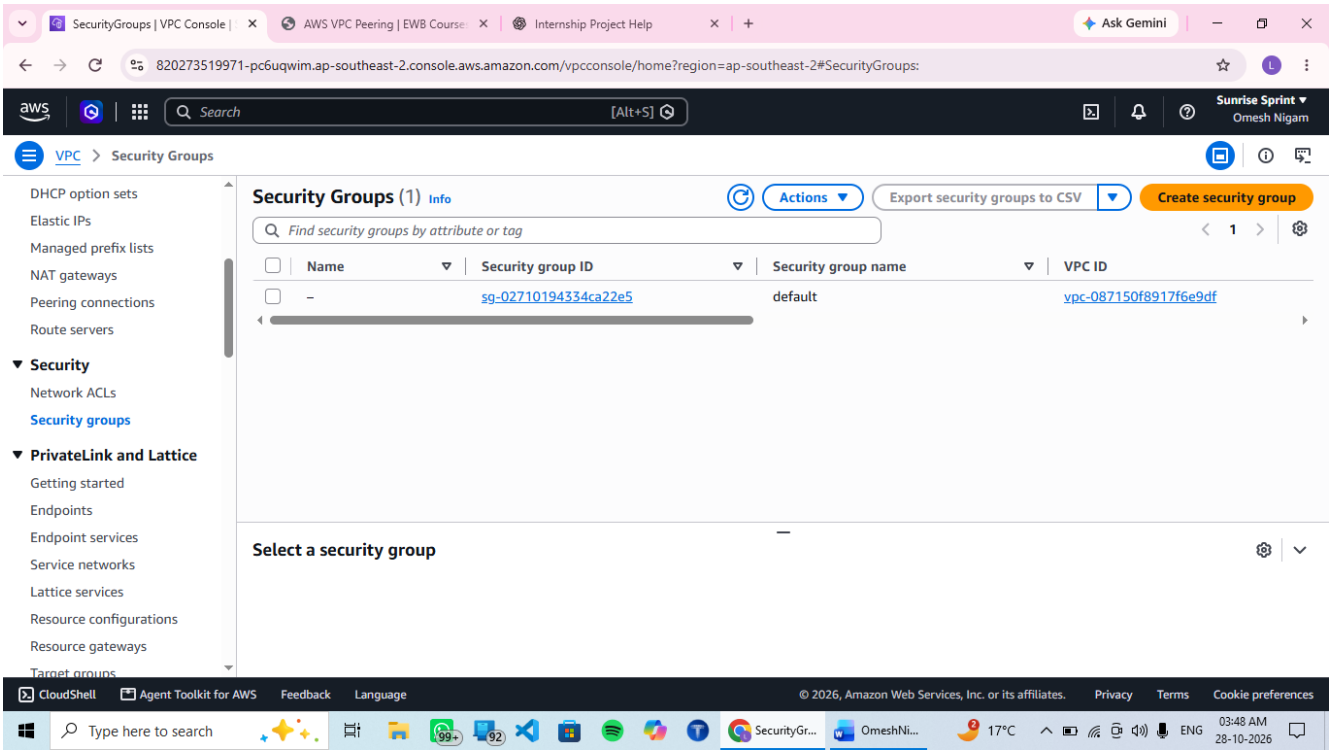



Subnets | VPC | Sunrise Sprint
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Ask Gemini
820273519971-pc6uqwim.ap-southeast-2.console.aws.amazon.com/vpcconsole/home?region=ap-southeast-2#subnets:
Search [Alt+S]
Sunrise Sprint
Omesh Nigam
VPC > Subnets
VPC dashboard
Filter by VPC
Choose a VPC
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Subnets (3) Info
Last updated 4 minutes ago
Actions
Create subnet
Find subnets by attribute or tag
Name
Subnet ID
State
VPC
Block Public.
subnet-036594b512f7f3546 Available vpc-087150f8917f6e9df Off
subnet-09a9a50e3fdd6035f Available vpc-087150f8917f6e9df Off
subnet-020bfec753ee8cb57 Available vpc-087150f8917f6e9df Off
Select a subnet
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Language
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Omesh Nigam
VPC > Route tables
VPC dashboard
Filter by VPC
Choose a VPC
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Route tables (1) Info
Last updated 4 minutes ago
Actions
Create route table
Find route tables by attribute or tag
Name
Route table ID
Explicit subnet associ...
Edge associations
Main
VPC
rtb-0eaa143428036dd7e – – Yes vpc-0
Select a route table
CloudShell
Agent Toolkit for AWS
Feedback
Language
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VPC > Internet gateways
VPC dashboard
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Choose a VPC
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Internet gateways (1) Info
Actions
Create internet gateway
Find internet gateways by attribute or tag
Name
Internet gateway ID
State
VPC ID
igw-0372bc3655175acc4
Attached
vpc-087150f8917f6e9df
Select an internet gateway above
CloudShell
Agent Toolkit for AWS
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VPC > Peering connections
Internet gateways
Egress-only internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs
Security groups
PrivateLink and Lattice
Getting started
Endpoints
Endpoint services
Service networks
Lattice services
Peering connections (1/1) Info
Actions
Create peering connection
Find peering connections by attribute or tag
Name
Peering connection ID
Status
Requester VPC
Accepter V
ewb-a-b-peer
pcx-0a3719dd486418f19
Deleted
vpc-0bc192b5a2485ca21 / ewb...
vpc-0f1d32
pcx-0a3719dd486418f19 / ewb-a-b-peer
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DNS
Route tables
Tags
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**Cleanup status:** Completed